

Subject: Discrete Mathematical Structures

Subject Code: BCS405A

Semester: 4th

Scheme: 2022

Module-2: Properties of the Integers

- ➔ **Mathematical Induction-well Ordered Principle**
- ➔ **Problems on Mathematical Induction**
- ➔ **Recursive Definition**
- ➔ **The Rules of Sum and Product**
- ➔ **Permutations**
- ➔ **Combinations**
- ➔ **Binomial and Multinomial Theorems**
- ➔ **Combination with Reputations**

website: vtucode.in

The method of mathematical induction is based on a Principle called the "Induction Principle". This Principle is based on another Principle known as "well-ordering Principle".

Note:- $\mathbb{Z}^+$ denotes the set of all positive integers.

Well-ordering Principle :-

OP "Every non-empty subset of $\mathbb{Z}^+$ contains a smallest (least) element"

(OR)

"The set of all +ve integers is well ordered".

OP Induction Principle :- (Principle of Mathematical Induction)

Let $S(n)$ denote an open statement that involves a +ve integer 'n'. Suppose that the foll. condns hold:

(i) $S(1)$ is true.

(ii) If whenever $S(k)$ is true for some particular, arbitrarily chosen $k \in \mathbb{Z}^+$, then $S(k+1)$ is true.

Then $S(n)$ is true for all $n \in \mathbb{Z}^+$.

Method of Mathematical Induction :-

Suppose we wish to prove that certain statement $S(n)$ is true, $\forall n \geq 1$. The method of proving this using principle of Induction is called the Method of Mathematical Induction. There are 2 steps in this method :-

1) Basic step : verify that $S(n)$ is true for $n=1$.

2) Induction step : Assuming $S(k)$ is true, where $k \geq 1$, show that $S(k+1)$ is true.

Note :-

* 1) without basic step, one cannot write the proof by induction.

✓ 2) Suppose we require to prove a statement $S(n)$ for $n \geq n_0$, where n_0 is an integer. In such situation, verifying $S(n_0)$ is true is the basic step, and in the Induction step, we take $k \geq n_0$.

Alternate form of Induction Principle :-

Let $S(n)$ denote a statement that involves a true integer 'n'. Suppose that

(i) for some true integers $n_0, n_0+1, n_0+2, \dots, n_0+p$, the statements $S(n_0), S(n_0+1), \dots, S(n_0+p)$ are true

and (ii) if $S(n_0), S(n_0+1), \dots, S(k)$ are true where $k \geq n_0+p$, then $S(k+1)$ is true.

Then $S(n)$ is true $\forall n \geq n_0$.

Problems

1) Prove by M.I that, $\forall n \geq 1, 1+2+3+4+\dots+n = \frac{1}{2}n(n+1)$ (2)

Soln:- let $S(n) : 1+2+3+\dots+n = \frac{1}{2}n(n+1)$

Basic step:- $S(1)$ is the statement

$$1 = \frac{1}{2} \cdot 1(2) \quad \text{which is true.}$$

Thus the statement $S(n)$ is verified for $n=1$.

Induction step:- Assume that $S(n)$ is true for $n=k$ where

$k \geq 1$.

i.e assume that $S(k) : 1+2+3+\dots+k = \frac{1}{2}k(k+1)$ is true

adding $(k+1)$ to both sides, we get

$$\begin{aligned} 1+2+3+\dots+k+k+1 &= \frac{1}{2}k(k+1) + (k+1) \\ &= \frac{1}{2}(k+1) [k+2] \\ &= \frac{1}{2}(k+1)(k+2) \end{aligned}$$

This is the statement $S(k+1)$.

Thus $S(k+1)$ is true, whenever $S(k)$ is true for $k \geq 1$.

Hence ~~the~~ proof by M.I, $S(n)$ is true $\forall n \geq 1$.

2) Prove by M.I, that $1^2+3^2+5^2+\dots+(2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$ $\forall n \geq 1$.

AP
Soln:- let $S(n) : 1^2+3^2+5^2+\dots+(2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$

Basic step: $S(1) : 1^2 = \frac{1}{3}(1)(1)(3)$ which is true.

Induction step: Assume that $S(n)$ is true for $n=k$, where

$$k \geq 1. \quad \text{i.e.} \quad 1^2+3^2+5^2+\dots+(2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$$

adding $(2k+1)^2$ on b.s

$$\begin{aligned} &[(k+1)-1]^2 \\ &= (2k+1)^2 \\ &= (2k+1)^2 \end{aligned}$$

$$\begin{aligned}
 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2k+1)^2 &= \frac{1}{3} k(2k-1)(2k+1) + (2k+1)^2 \\
 &= \frac{1}{3} (2k+1) [k(2k-1) + 3(2k+1)] \\
 &= \frac{1}{3} (2k+1) [2k^2 - k + 6k + 3] \\
 &= \frac{1}{3} (2k+1) (2k^2 + 5k + 3) \\
 &= \frac{1}{3} (2k+1)(k+1)(2k+3)
 \end{aligned}$$

$$\begin{aligned}
 &2k^2 + 3k + 2k + 3 \\
 &k(2k+3) + 1(2k+3) \\
 &(2k+3)(k+1)
 \end{aligned}$$

This is the statement $S(k+1)$.

Thus $S(k+1)$ is true whenever $S(k)$ is true, where $k \geq 1$.

Hence by M.I, $S(n)$ is true $\forall n \geq 1$.

3) Prove by M.I, $1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \dots + n(n+2) = \frac{n(n+1)(2n+7)}{6}$

Soln:- Let $S(n)$ denote the given statement.

Basic step:- $S(1)$: $1 \cdot 3 = \frac{1(1+1)(2 \cdot 1 + 7)}{6}$ which is true.

Induction step: Assume that $S(k)$ is true, where $k \geq 1$.

i.e. $1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \dots + k(k+2) = \frac{k(k+1)(2k+7)}{6}$

adding $(k+1)(k+3)$ on b.s, we get

$$1 \cdot 3 + 2 \cdot 4 + 3 \cdot 5 + \dots + k(k+2) + (k+1)(k+3) = \frac{k(k+1)(2k+7)}{6} + (k+1)(k+3)$$

$$\begin{aligned}
 &36 \quad 2k^2 + 9k + 4k + 18 \\
 &\wedge \quad k(2k+9) + 2(2k+9) \\
 &9 \quad 4 \quad (2k+9)(k+2)
 \end{aligned}$$

$$= \frac{(k+1)}{6} [k(2k+7) + 6(k+3)]$$

$$= \frac{(k+1)}{6} [2k^2 + 13k + 18]$$

$$= \frac{(k+1)(k+2)(2k+9)}{6}$$

This is the statement $S(k+1)$.

$\therefore S(k+1)$ is true whenever $S(k)$ is true, where $k \geq 1$.

Hence by M.I, $S(n)$ is true $\forall n \geq 1$.

Do 11th then
12th prob &
13th prob

4) Prove by M.I., for every the integer n , 5 divides $n^5 - n$ (3)

SP

Soln:- Note:- Ex:- 5 divides 30 means $\frac{30}{5}$.

(or) 30 is a multiple of 5

$$\underline{\text{i.e.}} \quad 30 = 5 \times m \quad (\text{Here } m=6) \quad \text{for some +ve integer } m$$

Let $S(n)$: 5 divides $n^5 - n$.

Basic step:- $S(1)$: 5 divides $1^5 - 1 = 0$. ($\underline{\text{i.e.}} \quad \frac{0}{5} = 0 \checkmark$)

which is true.

Induction step.. Assume that $S(n)$ is true for $n=k$ where

$k \geq 1$

$\underline{\text{i.e.}}$ for $k \geq 1$, 5 divides $k^5 - k$.

$\underline{\text{i.e.}}$ $k^5 - k$ is a multiple of 5

(or) $\boxed{k^5 - k = 5m}$, for some +ve integer 'm'.

$$\begin{aligned} \therefore (k+1)^5 - (k+1) &= (k+1)^2 (k+1)^2 (k+1) - (k+1) \\ &= (k^5 + 5k^4 + 10k^3 + 10k^2 + 5k + 1) - (k+1) \\ &= (k^5 - k) + 5(k^4 + 2k^3 + 2k^2 + k) \\ &= 5m + 5(k^4 + 2k^3 + 2k^2 + k) \\ &= 5(m + k^4 + 2k^3 + 2k^2 + k) \end{aligned}$$

$\Rightarrow (k+1)^5 - (k+1)$ is a multiple of 5

$\underline{\text{i.e.}}$ 5 divides $(k+1)^5 - (k+1)$. This is precisely the $S(k+1)$.

Thus $S(k+1)$ is true whenever $S(k)$ is true for $k \geq 1$.

$\therefore$ By M.I., $S(n)$ is true $\forall n \geq 1$.

5) P.T $4n < (n^2 - 7) \quad \forall$ +ve integers $n \geq 6$.

SP
Soln:- Let $S(n)$: $4n < (n^2 - 7)$.

To prove that $S(n)$ is true $\forall$ +ve integers $n \geq n_0$, where $n_0 = 6$.

Basic step : $S(6) : (4 \times 6) < (6^2 - 7)$

$$\text{ie } 24 < 29 \text{ which is true}$$

$\therefore S(n)$ is true for $n = n_0 = 6$.

Induction step : Assume that $S(n)$ is true for $n = k$ where

$$k \geq 6 \quad \text{ie} \quad 4k < (k^2 - 7) \quad \text{for } k \geq 6.$$

add 4 on b.s

$$\text{then } 4(k+1) = 4k + 4 < (k^2 - 7) + 4$$
$$< (k^2 - 7) + (2k + 1)$$

because when $k \geq 6$, we have $2k \geq 12$

$$\Rightarrow 2k + 1 \geq 13 > 4$$

$$\therefore 4(k+1) < (k^2 + 2k + 1) - 7$$

$$\text{ie } 4(k+1) < (k+1)^2 - 7 \quad \text{This is the statement } S(k+1)$$

$\Rightarrow S(k+1)$ is true, whenever $S(k)$ is true $\forall k \geq 6$.

Thus $S(n)$ is true $\forall n \geq 6$.

6) Show that $2^n > n^2$ $\forall$ the integers 'n' greater than 4.

Soln :- To prove $S(n) : 2^n > n^2$ is true $\forall n \geq 5$

Basic step : for $n = 5$, $S(n)$ gives

$$S(5) : 2^5 > 5^2 \quad (\text{ie } 32 > 25) \text{ which is true}$$

Induction step : Assume that $S(n)$ is true for $n = k$, where

$$k \geq 5 \quad \text{ie} \quad 2^k > k^2 \quad \text{for } k \geq 5.$$

Multiplying by 2 on b.s

$$2 \cdot 2^k = 2^{k+1} > 2k^2$$

Since $k > 4$, we have

$$2k^2 \geq k^2 + k^2 = k^2 + (k \times k)$$

$$> k^2 + 4k$$

$$> k^2 + 2k + 2k$$

$$> k^2 + 2k + 1 = (k+1)^2$$

thus $2^{k+1} > (k+1)^2$ for $k \geq 4$

(4)

This is precisely the statement $S(k+1)$.

Thus $S(k+1)$ is true whenever $S(k)$ is true $\forall k \geq 4$.

Hence by M.I., $S(n)$ is true $\forall n \geq 4$.

7) By M.I., prove that $n! \geq 2^{n-1} \forall n \geq 1$.

Soln:- Let $S(n)$ be the given statement.

Basic step: for $n=1$, $S(1)$: $1! \geq 2^{1-1}$
 $1 \geq 1$ which is true.

Induction step: Assume that $S(n)$ is true for $n=k$,
 where $k \geq 1$. i.e. $k! \geq 2^{k-1}$ for $k \geq 1$.

i.e. $2^{k-1} \leq k!$

$2^k = 2^{k-1} \cdot 2$ multiplying by 2 on b.s

$2^k = 2 \cdot 2^{k-1} \leq 2 \cdot k!$

$\leq (k+1) \cdot k!$ because $(k+1) \geq 2$ for $k \geq 1$
 $= (k+1)!$

i.e. $(k+1)! \geq 2^k$, which is the statement $S(k+1)$.

Thus $S(k+1)$ is true whenever $S(k)$ is true for $k \geq 1$.

Hence by M.I., $S(n)$ is true $\forall n \geq 1$.

8) for all $n \in \mathbb{Z}^+$, show that if $n \geq 14$, then n can be written as a sum of 3's and/or 8's.

P.T.O.

Soln: To prove:

$S(n)$ n can be written as a sum of 3's or 8's
is true for all integers $n \geq 14$.

Basic step: we have $14 = (3+3)+8$.

$\Rightarrow S(14)$ is true.

Induction step: Assume that $S(n)$ is true for $n = k$ where $k \geq 14$.

then $k = (3+3+3+\dots) + (8+8+\dots)$

suppose this representation of k has r no of 3's and s no of 8's.

Since $k \geq 14$, we must have $r \geq 2$, $s \geq 1$.

$$\begin{aligned} \therefore k+1 &= \underbrace{(3+3+3+\dots)}_r + \underbrace{(8+8+\dots)}_s + 1 \\ &= \underbrace{(3+3+3+\dots)}_r + \underbrace{(8+8+\dots)}_{s-1} + 8+1 \end{aligned}$$

$$= \underbrace{(3+3+3+\dots)}_r + \underbrace{(8+8+\dots)}_{s-1} + (3+3+3)$$

$$= \underbrace{(3+3+3+\dots)}_{r+3} + \underbrace{(8+8+\dots)}_{s-1}$$

$\Rightarrow (k+1)$ is a sum of 3's & 8's. i.e. $(k+1)$ is true.

Hence by H.I, $S(n)$ is true for all the integers $n \geq 14$.

Q) By H.I, Prove that, for every the integer n , the no
 $A_n = 5^n + 2 \cdot 3^{n-1} + 1$ is a multiple of 8.

Soln:- Let $S(n)$ be the given statement.

Basic step: $S(1): 5^1 + 2 \cdot 3^0 + 1 = 8$ is a multiple of 8.

$\therefore$ for $n=1$, A_n is ~~(a multiple of 8)~~ true.

Induction step: Assume that a_n is a multiple of 8 for $n = k \geq 1$. (5)

i.e. $A_k = 5^k + 2 \cdot 3^{k-1} + 1$ is a multiple of 8

we find,

$$\begin{aligned} A_{k+1} - A_k &= (5^{k+1} + 2 \cdot 3^k + 1) - (5^k + 2 \cdot 3^{k-1} + 1) \\ &= (5^k \cdot 5 - 5^k) + 2(3^k - 3^{k-1}) \\ &= 5^k(5 - 1) + 2 \cdot 3^{k-1}(3 - 1) \\ &= 4 \cdot 5^k + 4 \cdot 3^{k-1} \\ &= 4(5^k + 3^{k-1}) \end{aligned}$$

5 & 3 are odd $\Rightarrow 5^k$ & 3^{k-1} are also odd.

$\Rightarrow 5^k + 3^{k-1}$ is even.

$\Rightarrow 4(5^k + 3^{k-1})$ is a multiple of 8.

i.e. $A_{k+1} - A_k$ is a multiple of 8.

Since A_k is a multiple of 8 (by assumption), A_{k+1} is also a multiple of 8.

if A_n is a multiple of 8 for $n = k$, then

Thus A_n is a multiple of 8 for $n = k+1$.

Hence by M.I., for every ^{tr} integer n , $A_n = 5^n + 2 \cdot 3^{n-1} + 1$ is a multiple of 8.

10) Prove by mathematical Induction that $6^{n+2} + 7^{2n+1}$ is divisible by 43 for each ^{tr} integer n .

Soln:- Let $A_n = 6^{n+2} + 7^{2n+1}$

Basic step: $A_1 = 6^3 + 7^3 = 559$ is divisible by 43.

$\therefore A_n$ is divisible by 43 for $n=1$.

Induction step: Assume that A_n is divisible by 43

$$\text{for } n = k \geq 1 \quad \text{ie } A_k = 6^{k+2} + 7^{2k+1}$$

$$\begin{aligned} \therefore A_{k+1} &= 6^{k+3} + 7^{2(k+1)+1} \\ &= 6^{k+3} + 7^{2k+3} \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 7^2 \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 49 \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} (6 + 43) \\ &= 6^{k+2} \cdot 6 + 7^{2k+1} \cdot 6 + 7^{2k+1} \cdot 43 \\ &= 6 [6^{k+2} + 7^{2k+1}] + 43 \cdot 7^{2k+1} \\ &= 6A_k + 43 \cdot 7^{2k+1} \end{aligned}$$

By assumption, A_k is divisible by 43 $\Rightarrow 6A_k$ is also divisible by 43, and $43 \cdot 7^{2k+1}$ is divisible by 43

$\therefore 6A_k + 43 \cdot 7^{2k+1}$ is divisible by 43

$\therefore A_{k+1}$ is divisible by 43.

2) A_{k+1} is divisible by 43. Thus if $S(k)$ is true, then $S(k+1)$ is true for $k \geq 1$.

Hence by M.I, $S(n)$ is true $\forall n \geq 1$.

11) Prove $\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1} \quad \forall n \in \mathbb{Z}.$

$$\text{ie } \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

Soln:- Let $S(n)$ be the given statement.

Basic step: $S(1): \frac{1}{1(2)} = \frac{1}{1+1} \Rightarrow \frac{1}{2} = \frac{1}{2}$ which is true

P.T.O

Induction step: Assume $S(k)$ is true for $k \geq 1$

(6)

i.e. $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$ is true.

adding $\frac{1}{(k+1)(k+2)}$ on b.s

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{(k+1)(k+2)} = \frac{k}{k+1} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{1}{k+1} \left[k + \frac{1}{k+2} \right]$$

$$= \frac{1}{(k+1)} \left[\frac{k^2 + 2k + 1}{k+2} \right]$$

$$= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2}$$

which is precisely the statement $S(k+1)$.

Thus $S(k+1)$ is true, whenever $S(k)$ is true.

Hence by H.I, $S(n)$ is true $\forall n \geq 1$.

12) Establish the following by H.I

Q.P.

$$\sum_{i=1}^n i 2^i = 2 + (n-1) 2^{n+1}$$

Soln: Let $S(n)$ be the given statement.

Basic step: $S(1)$: $1 \times 2^1 = 2 + (1-1) 2^2$

$2 = 2$ which is true.

Induction step: Assume that the result is true for $n = k$,

where $k \geq 1$.

i.e. $S(k)$: $\sum_{i=1}^k i 2^i = 2 + (k-1) 2^{k+1}$ is true.

i.e. $1 \times 2^1 + 2 \times 2^2 + 3 \times 2^3 + 4 \times 2^4 + \dots + k \times 2^k = 2 + (k-1) 2^{k+1}$

add $(k+1) \times 2^{k+1}$ on b.s

$$1 \times 2^1 + 2 \times 2^2 + 3 \times 2^3 + \dots + k \times 2^k + (k+1) \times 2^{(k+1)}$$

$$= 2 + (k-1) 2^{k+1} + (k+1) \times 2^{(k+1)}$$

$$= 2 + 2^{k+1} [k-1 + k+1]$$

$$= 2 + 2^{k+1} \cdot 2k$$

$$= 2 + k \cdot 2^{(k+2)}$$

which is precisely the statement $S(k+1)$.

Thus $S(k+1)$ is true whenever $S(k)$ is true, $\forall k \geq 1$.

Hence $S(n)$ is true $\forall n \geq 1$.

Hw

1) P.T $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6} n(2n+1)(n+1)$

2) P.T $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$

3) P.T $2^n < n!$ $\forall$ the integers $n > 2$.

4) for every integer n , 3 divides $n^3 - n$.

5) P.T for all $n \in \mathbb{Z}^+$, s.t. if $n \geq 24$ then n can be written as a sum of 5's and 7's.

13) Let $H_1 = 1$, $H_2 = 1 + \frac{1}{2}$, $H_3 = 1 + \frac{1}{2} + \frac{1}{3}$, $\dots$, $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$

P.T Prove that $\sum_{i=1}^n H_i = (n+1)H_n - n$, $\forall n \geq 1$.

Soln: Let $S(n) : \sum_{i=1}^n H_i = (n+1)H_n - n$, $\forall n \geq 1$.
 $\Rightarrow H_1 + H_2 + H_3 + \dots + H_n = (n+1)H_n - n$.

Basic step: $S(1) : H_1 = (1+1)H_1 - 1$.

$$1 = 2(1) - 1.$$

$$1 = 1.$$

$\therefore S(1)$ is true.

Induction step: Assume $S(k)$ is true $\forall k \geq 1$.

i.e $S(k) : \sum_{i=1}^k H_i = (k+1)H_k - k$ is true $\forall k \geq 1$.

i.e $S(k) : H_1 + H_2 + \dots + H_k = (k+1)H_k - k$ is true $\forall k \geq 1$.

adding H_{k+1} on b.s.

$$H_1 + H_2 + \dots + H_k + H_{k+1} = (k+1)H_k - k + H_{k+1}$$

$$= (k+1) \left\{ H_{k+1} - \frac{1}{k+1} \right\} - k + H_{k+1}$$

$$\left[\because H_k = H_{k+1} - \frac{1}{(k+1)} \text{ from the defn of } H_n \right]$$

$$= (k+1) H_{k+1} - 1 - k + H_{k+1}$$

$$= (k+1+1) H_{k+1} - (k+1)$$

$$= (k+2) H_{k+1} - (k+1)$$

this is precisely the statement $S(k+1)$.

By M.I, $S(n)$ is true $\forall n \geq 1$.

Recursive Definitions

Consider $a_1, a_2, \dots, a_n$, a sequence in which a_1 is the first term, a_2 is the 2nd term, $\dots$, a_n is n^{th} term. The element a_n is called the general term of the sequence denoted by $\{a_n\}$. If the first term is denoted by a_0 , then n^{th} term is denoted by a_{n-1} and this sequence is denoted by $\{a_{n-1}\}$.

For describing a sequence, 2 methods are commonly used:
(i) Explicit method (ii) Recursive method.

In Explicit method, the general term of the sequence is explicitly indicated.

Ex:- $E = \{a_n\}$, where $a_n = 2n$, $n \in \mathbb{Z}^+$.

In Recursive method, first few terms of the sequence are explicitly indicated and the general term is specified through a rule (formula) which indicates how to obtain new terms of the sequence from the terms already known.

Ex:- fibonacci numbers $f_0 = 0$, $f_1 = 1$, $f_n = f_{n-1} + f_{n-2} \quad \forall n \geq 2$.

Note :- 1) A recursive definition of a sequence should contain 2 parts: in the 1st part, a few terms of the sequence are indicated explicitly and in the 2nd part, the rule to obtain new terms for sequence is indicated.

2) Recursive rule/formula: is a formula for a_n in terms of a_{n-1} , a_{n-2} etc (or) a formula for a_{n+1} in terms of a_n , a_{n-1} etc.

P.T.O.

Problems

1) obtain a recursive defⁿ for the sequence $\{a_n\}$ in each of the following cases:

i) $a_n = 6^n$

ii) $a_n = n(n+3)$

iii) $a_n = (n+1)!$

iv) $a_n = 2 - (-1)^n$

Soln:- i) $a_n = 6^n$

$$a_1 = 6^1 = 6$$

$$a_2 = 6^2 = 36 = 6 \times 6 = 6 \times a_1$$

$$a_3 = 6^3 = 216 = 6 \times 36 = 6 \times a_2$$

$$a_4 = 6^4 = 1296 = 6 \times 216 = 6 \times a_3$$

$\vdots$

$$a_1 = 6; \boxed{a_n = 6 a_{n-1}} \quad \forall n \geq 2 \text{ is the recursive def.}$$

$\rightarrow$ same as $a_{n+1} = 6 a_n \quad \forall n \geq 1$

ii) $a_n = n(n+3)$

$$a_1 = 1(1+3) = 1 \times 4 = 4$$

$$a_2 = 2(2+3) = 2 \times 5 = 10 = a_1 + 6 = a_1 + (2 \times 3)$$

$$a_3 = 3(3+3) = 3 \times 6 = 18 = a_2 + 8 = a_2 + (2 \times 4)$$

$$a_4 = 4(4+3) = 4 \times 7 = 28 = a_3 + 10 = a_3 + (2 \times 5)$$

$\vdots$

$$a_n = a_{n-1} + 2(n+1) \quad \forall n \geq 2$$

Thus $a_1 = 4, a_n = a_{n-1} + 2(n+1) \quad \forall n \geq 2$ is the recursive defⁿ.

iii) $a_n = (n+1)!$

$$a_1 = (1+1)! = 2! = 2$$

$$a_2 = (2+1)! = 3! = 6 = a_1 \times 3$$

$$a_3 = (3+1)! = 4! = 24 = a_2 \times 4$$

$$a_4 = (4+1)! = 5! = 120 = a_3 \times 5$$

$$a_n = a_{n-1} \times (n+1) \quad \forall n \geq 2$$

Thus $a_1 = 2, a_n = (n+1) a_{n-1} \quad \forall n \geq 2$ is the recursive defⁿ.

iv) $a_n = 2 - (-1)^n$

$$a_1 = 2 - (-1)^1 = 2 + 1 = 3$$

$$a_2 = 2 - (-1)^2 = 2 - 1 = 1 = 3 - 2 = a_1 - (-1)^2 \times 2$$

$$a_3 = 2 - (-1)^3 = 2 + 1 = 3 = 1 + 2 = a_2 - (-1)^3 \times 2$$

$$a_4 = 2 - (-1)^4 = 2 - 1 = 1 = 3 - 2 = a_3 - (-1)^4 \times 2$$

$$a_5 = 2 - (-1)^5 = 2 + 1 = 3$$

$$a_n = a_{n-1} - 2(-1)^n \quad \forall n \geq 2 \text{ is the recursive defⁿ}$$

$$a_1 = 3$$

2) Find an explicit defⁿ of the sequence defined recursively by
 $a_1 = 7, a_n = 2a_{n-1} + 1$ for $n \geq 2$

Soln:- $a_n = 2a_{n-1} + 1$

$$= 2^1 a_{n-1} + (2^1 - 1)$$

$$= 2 \{ 2a_{n-2} + 1 \} + 1 = 4a_{n-2} + 3 = 2^2 a_{n-2} + (2^2 - 1)$$

$$= 4 \{ 2a_{n-3} + 1 \} + 3 = 8a_{n-3} + 7 = 2^3 a_{n-3} + (2^3 - 1)$$

$$= 8 \{ 2a_{n-4} + 1 \} + 7 = 16a_{n-4} + 15 = 2^4 a_{n-4} + (2^4 - 1)$$

$$\vdots$$

$$a_n = 2^{n-1} \cdot a_1 + 2^{n-1} - 1$$

$$= 2^{n-1} \{ a_1 + 1 \} - 1$$

$$= 2^{n-1} \{ 7 + 1 \} - 1$$

$$= 2^{n-1} (8) - 1$$

$$= 2^{n-1} \cdot 2^3 - 1 = 2^{n+2} - 1$$

$$= 4 \times 2^n - 1$$

Thus $a_n = 4 \times 2^n - 1$ is an explicit defⁿ for the given sequence.

3) A sequence $\{a_n\}$ is defined recursively by $a_1 = 4, a_n = a_{n-1} + n$
 $\forall n \geq 2$. find a_n in explicit form.

Soln:- $a_n = a_{n-1} + n$

$$= \{ a_{n-2} + (n-1) \} + n$$

$$= a_{n-3} + (n-2) + (n-1) + n$$

$$= a_{n-4} + (n-3) + (n-2) + (n-1) + n$$

taking $a_{n-4} = a_1$
 $\Rightarrow n-4 = 1 \Rightarrow n = 5$

$$= a_1 + 2 + 3 + 4 + \dots + n$$

$$= (a_1 - 1) + (1 + 2 + 3 + \dots + n)$$

$$= (a_1 - 1) + \frac{n(n+1)}{2}$$

$$\boxed{a_n = 3 + \frac{n(n+1)}{2}}$$
 is the explicit formula for a_n .

4) A sequence $\{a_n\}$ is defined recursively by

$a_0 = 1, a_1 = 1, a_2 = 1$ and $a_n = a_{n-1} + a_{n-3} \quad \forall n \geq 3$.

Prove that $a_{n+2} > (\sqrt{2})^n \quad \forall n \geq 0$.

Soln: Let $S(n) : a_{n+2} > (\sqrt{2})^n$

Basic step: $S(0) : a_2 > (\sqrt{2})^0$

$$1 > 1 - \text{True}$$

$S(1) : a_3 > (\sqrt{2})^1$

$$2 > \sqrt{2} - \text{True}$$

$S(2) : a_4 > (\sqrt{2})^2$

$$2 > 2$$

$$(a_3 = ?$$

$$a_n = a_{n-1} + a_{n-3}$$

$$a_3 = a_2 + a_0 = 1 + 1 = 2$$

$$a_4 = a_3 + a_1 = 1 + 1 = 2$$

Thus $S(n)$ is true for $n = 0, 1, 2$.

Induction step: Assume $S(k)$ is true

i.e. $S(k) : a_{k+2} > (\sqrt{2})^k \quad \forall k \geq 0$ is true

TPT: $a_{k+3} > (\sqrt{2})^{k+1}$

Consider $a_{k+3} = a_{k+2} + a_k$ (using the def of a_n)

$$\geq (\sqrt{2})^k + (\sqrt{2})^{k-2} \quad (\text{using } \textcircled{1})$$

$$> (\sqrt{2})^{k-2} [(\sqrt{2})^2 + 1]$$

$$> (\sqrt{2})^{k-2} [3]$$

$$> (\sqrt{2})^{k+1-1-2} (3)$$

$$> (\sqrt{2})^{(k+1)} \cdot (\sqrt{2})^{-3} (3)$$

$$> (\sqrt{2})^{(k+1)} \frac{3}{2\sqrt{2}}$$

$$> (\sqrt{2})^{(k+1)} \quad \left(\because \frac{3}{2\sqrt{2}} > 1 \right)$$

Thus $a_{k+3} > (\sqrt{2})^{k+1}$

Hence by M.I., required result is true $\forall n \geq 0$.

5) For the sequence $\{a_n\}$ defined recursively by

$a_1 = 8, a_2 = 22, a_n = 4(a_{n-1} - a_{n-2})$ for $n \geq 3$, prove that

$$a_n = (5 + 3n)2^{n-1} \quad \text{for } n \geq 1.$$

Soln :- $S(n) : a_n = (5+3n) 2^{n-1} \quad \forall n \geq 1$

Basic step : $S(1) : a_1 = (5+3) 2^0 = 8$

$S(2) : a_2 = (5+6) 2 = 22$

$S(n)$ is true for $n=1, 2$.

Induction step : Assume $S(k) : a_k = (5+3k) 2^{k-1}$ is true $\forall k \geq 1$
 $\hookrightarrow$ (1)

TPT : $a_{k+1} = \{5+3(k+1)\} 2^{k+1-1}$

i.e. $a_{k+1} = \{8+3k\} 2^k$.

Consider $a_{k+1} = 4(a_k - a_{k-1})$ (using the defn of a_n)

$a_{k+1} = 4 \left[(5+3k) 2^{k-1} - (5+3(k-1)) 2^{k-2} \right]$ (using (1))

$= 4 \left[5(2^{k-1} - 2^{k-2}) + 3 \{ k \cdot 2^{k-1} - (k-1) 2^{k-2} \} \right]$

$= 4 \left[5 \cdot 2^{k-2} (2-1) + 3 \cdot 2^{k-2} \{ 2k - k + 1 \} \right]$

$= 4 \left[5 \cdot 2^{k-2} + 3 \cdot 2^{k-2} (k+1) \right]$

$= 4 \cdot 2^{k-2} [5 + 3k + 3]$

$= 4 \cdot 2^k \cdot 2^{-2} [8 + 3k]$

$= 2^k (8+3k)$ is the statement $S(k+1)$.

Thus by M.I, required result is true $\forall n \geq 1$.

6) A sequence $\{C_n\}$ is defined recursively by $C_n = 3C_{n-1} - 2C_{n-2}$
 $\forall n \geq 3$, with $C_1 = 5$ and $C_2 = 3$. S.T. $C_n = -2^n + 7$.

Soln :- Let $S(n) : C_n = -2^n + 7 \quad \forall n \geq 1$.

Basic step : $S(1) : C_1 = -2^1 + 7$
 $5 = 5$ - true.

$S(2) : C_2 = -2^2 + 7$
 $3 = 3$ - true

$\therefore S(n)$ is true for $n=1, 2$.

4) Induction step: Assume $S(k)$: $C_k = -2^k + 7$ is true $\forall k \geq 1$

Q.P. T.P.T: $S(k+1)$: $C_{k+1} = -2^{k+1} + 7$

Sol Consider $C_{k+1} = 3C_k - 2C_{k-1}$ (by the defⁿ of C_n)

$$= 3(-2^k + 7) - 2(-2^{k-1} + 7)$$

$$= (-3 \cdot 2^k + 2 \cdot 2^{k-1}) + 21 - 14$$

$$= 2^{k-1}(-3 \cdot 2 + 2) + 7$$

$$= -4 \cdot 2^{k-1} + 7$$

$$= -2^2 \cdot 2^{k-1} + 7$$

$$= -2^{k+1} + 7$$

which is exactly the statement $S(k+1)$.

Thus by M.I, $S(n)$ is true $\forall n \geq 1$.

7) The Fibonacci numbers are defined by $f_0 = 0, f_1 = 1$ and
Q.P. $f_n = f_{n-1} + f_{n-2} \forall n \geq 2$. Prove that $\sum_{i=0}^n f_i^2 = f_n \times f_{n+1}$

Soln: Let $S(n)$: $\sum_{i=0}^n f_i^2 = f_n \times f_{n+1}$

Basic step: $S(0)$: $f_0^2 = f_0 \times f_1$
 $0^2 = 0 \times 1$
 $0 = 0$

$S(1)$: $f_1^2 = f_1 \times f_2$
 $1^2 = 1 \times (f_1 + f_0)$
 $1 = 1 \times (1 + 0)$
 $1 = 1$

$S(n)$ is true for $n = 0, 1$.

Induction step: Assume $S(k)$ is true $\forall k \geq 1$.

i.e. $S(k)$: $\sum_{i=0}^k f_i^2 = f_k \times f_{k+1} \forall k \geq 1$ $\rightarrow$ ①

T.P.T: $S(k+1)$: $\sum_{i=0}^{k+1} f_i^2 = f_{k+1} \times f_{k+2}$

Consider $\sum_{i=0}^{k+1} f_i^2 = f_0^2 + f_1^2 + \dots + f_k^2 + f_{k+1}^2$

$$\begin{aligned}
 \sum_{i=0}^{k+1} F_i^2 &= \sum_{i=0}^k F_i^2 + F_{k+1}^2 \\
 &= (F_k \times F_{k+1}) + F_{k+1}^2 \quad (\text{using the 4th } \textcircled{1}) \\
 &= F_{k+1} (F_k + F_{k+1}) \\
 &= F_{k+1} (F_{k+2}) \quad (\because F_n = F_{n-1} + F_{n-2}) \\
 &= F_{k+1} \times F_{k+2}
 \end{aligned}$$

this is exactly the statement $S(k+1)$.

Thus by M.I, $S(n)$ is true $\forall n \geq 0$.

g) for the fibonacci sequence, $F_0, F_1, F_2, \dots$ prove that

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$$

Soln:- Let $S(n): F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right]$

Basic step: $S(0): F_0 = \frac{1}{\sqrt{5}} [1 - 1] = 0$.

$$\begin{aligned}
 S(1): F_1 &= \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right) - \left(\frac{1-\sqrt{5}}{2} \right) \right] \\
 &= \frac{1}{\sqrt{5}} \left[\frac{1}{2} + \frac{\sqrt{5}}{2} - \frac{1}{2} + \frac{\sqrt{5}}{2} \right] \\
 &= \frac{1}{\sqrt{5}} \cdot \frac{2\sqrt{5}}{2} = 1.
 \end{aligned}$$

Thus $S(n)$ is true for $n=0, 1$.

Induction step: Assume $S(k)$ is true $\forall k \geq 1$.

i.e. $S(k): F_k = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^k - \left(\frac{1-\sqrt{5}}{2} \right)^k \right] \quad \forall k \geq 1$.

TPP: $S(k+1)$ is true i.e. $F_{k+1} = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{k+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{k+1} \right]$.

Consider $F_{k+1} = F_k + F_{k-1}$ (by defⁿ of fibonacci no's)

$$\begin{aligned}
 \Rightarrow F_{k+1} &= \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^k - \left(\frac{1-\sqrt{5}}{2} \right)^k \right] + \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{k-1} - \left(\frac{1-\sqrt{5}}{2} \right)^{k-1} \right] \\
 &= \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{k-1} \left\{ \frac{1+\sqrt{5}}{2} + 1 \right\} - \left(\frac{1-\sqrt{5}}{2} \right)^{k-1} \left\{ \frac{1-\sqrt{5}}{2} + 1 \right\} \right]
 \end{aligned}$$

$$\begin{aligned}
 f_{k+1} &= \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{k+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{k+1} \right] \\
 &= \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^k \left(\frac{1+\sqrt{5}}{2} \right) - \left(\frac{1-\sqrt{5}}{2} \right)^k \left(\frac{1-\sqrt{5}}{2} \right) \right] \\
 &\left[\because \left(\frac{1+\sqrt{5}}{2} \right)^2 = \frac{1+5+2\sqrt{5}}{4} = \frac{6+2\sqrt{5}}{4} = \frac{2(3+\sqrt{5})}{4} = \frac{3+\sqrt{5}}{2} \right]
 \end{aligned}$$

$$\therefore f_{k+1} = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^{k+1} - \left(\frac{1-\sqrt{5}}{2} \right)^{k+1} \right]$$

Thus $S(k+1)$ is true whenever $S(k)$ is true.

Thus by M.I, the result is true $\forall n \geq 0$.

9) Let f_n denotes the fibonacci numbers. Prove that

Q.P
$$\sum_{i=1}^n \frac{f_{i-1}}{2^i} = 1 - \frac{f_{n+2}}{2^n}$$

Soln:- Let $S(n): \sum_{i=1}^n \frac{f_{i-1}}{2^i} = 1 - \frac{f_{n+2}}{2^n}$

Basic step: $n=1$; $S(1): \frac{f_0}{2^1} = 1 - \frac{f_3}{2^1}$

we have $f_0 = 0, f_1 = 1, f_2 = f_1 + f_0 = 1$; $f_3 = f_2 + f_1 = 1 + 1 = 2$.

$\therefore S(1): \frac{0}{2} = 1 - \frac{2}{2} \Rightarrow 0 = 0$

$\therefore S(n)$ is true for $n=1$.

Induction step: Assume $S(k)$ is true $\forall k \geq 1$.

i.e $S(k): \sum_{i=1}^k \frac{f_{i-1}}{2^i} = 1 - \frac{f_{k+2}}{2^k} \rightarrow \text{①}$

TPT: $S(k+1): \sum_{i=1}^{k+1} \frac{f_{i-1}}{2^i} = 1 - \frac{f_{k+3}}{2^{k+1}}$

Consider
$$\begin{aligned}
 \sum_{i=1}^{k+1} \frac{f_{i-1}}{2^i} &= \frac{f_0}{2} + \frac{f_1}{2^2} + \frac{f_2}{2^3} + \dots + \frac{f_{k-1}}{2^k} + \frac{f_k}{2^{k+1}} \\
 &= \underbrace{\sum_{i=1}^k \frac{f_{i-1}}{2^i}}_{\text{①}} + \frac{f_k}{2^{k+1}}
 \end{aligned}$$

$$\Rightarrow \sum_{i=1}^{k+1} \frac{f_{i-1}}{2^i} = \left(1 - \frac{f_{k+2}}{2^k}\right) + \frac{f_k}{2^{k+1}} \quad (\text{using } \textcircled{1})$$

$$= 1 - \frac{1}{2^k} \left[f_{k+2} - \frac{f_k}{2} \right]$$

$$= 1 - \frac{1}{2^k} \left[\frac{2f_{k+2} - f_k}{2} \right]$$

$$= 1 - \frac{1}{2^{k+1}} \left[f_{k+2} + f_{k+2} - f_k \right]$$

$$\text{but } f_{k+2} = f_{k+1} + f_k$$

$$\Rightarrow f_{k+2} - f_k = f_{k+1}$$

$$\therefore \sum_{i=1}^{k+1} \frac{f_{i-1}}{2^i} = 1 - \frac{1}{2^{k+1}} \left[f_{k+2} + f_{k+1} \right]$$

$$= 1 - \frac{1}{2^{k+1}} f_{k+3} \quad \left(\because f_n = f_{n-1} + f_{n-2} \right. \\ \left. f_{k+3} = f_{k+2} + f_{k+1} \right)$$

this is the statement $S(k+1)$.

Thus by m.I, the required result is true $\forall n \geq 1$.

10) If f_i 's are fibonacci numbers defined recursively by $f_0 = 0, f_1 = 1$ and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$ and L_i 's are

Lucas numbers defined recursively as $L_0 = 2, L_1 = 1$ and $L_n = L_{n-1} + L_{n-2} \forall n \geq 2$, then P.T $L_n = f_{n-1} + f_{n+1} \forall n \geq 1$.

Soln:- Let $S(n): L_n = f_{n-1} + f_{n+1} \forall n \geq 1$.

Basic Step: $S(1): L_1 = f_0 + f_2$
 $1 = 0 + 1 = 1 \quad (\because f_2 = f_1 + f_0 = 1 + 0 = 1)$

$\therefore S(n)$ is true for $n=1$.

Induction step: Assume $S(k)$ is true $\forall k \geq 1$.

$$\therefore L_k = f_{k-1} + f_{k+1} \quad \forall k \geq 1 \rightarrow \textcircled{1}$$

$$\text{TPT: } S(k+1): L_{k+1} = f_k + f_{k+2}$$

Consider $L_{k+1} = L_k + L_{k-1}$ (by the def of L_n)

$$\begin{aligned} \Rightarrow L_{k+1} &= (f_{k-1} + f_{k+1}) + (f_{k-2} + f_k) \quad (\text{by the assumption made}) \\ &= (f_{k-1} + f_{k-2}) + (f_{k+1} + f_k) \\ &= f_k + f_{k+2} \quad (\text{by def of } f_n) \end{aligned}$$

this is the statement $S(k+1)$

Thus by m.I, the required result is true $\forall n \geq 1$

11) The Ackerman's numbers are defined recursively as follows:

$$A_{0,n} = n+1 \quad \text{for } n \geq 0 \rightarrow (1)$$

$$A_{m,0} = A_{m-1,1} \quad \text{for } m \geq 0 \rightarrow (2)$$

$$A_{m,n} = A_{m-1,p} \quad \text{where } p = A_{m,n-1} \quad \text{for } m \geq 1, n \geq 1 \rightarrow (3)$$

Prove that $A_{1,n} = n+2 \quad \forall n \in \mathbb{N}$.

Soln: Let $S(n): A_{1,n} = n+2 \quad \forall n \in \mathbb{N}$

Basic step: $S(1): A_{1,1} = 1+2 \Rightarrow \text{RHS} = 3$

$$\begin{aligned} \text{consider LHS: } A_{1,1} &= A_{0,1} \quad (\text{by } (2)) \\ A_{1,1} &= 1+1 = 2 \quad (\text{by } (1)) \end{aligned}$$

consider LHS: $A_{1,1} = A_{0,p}$ where $p = A_{1,0} = A_{0,1} = 2$ (by (2) & (1))

$$\begin{aligned} \therefore A_{1,1} &= A_{0,2} \\ &= 2+1 \quad (\text{by } (1)) \end{aligned}$$

$$\text{LHS} = 3$$

$\therefore S(1)$ is true.

Induction step: Assume $S(k)$ is true $\forall k \geq 1$.

$$\text{ii } A_{1,k} = k+2 \quad \forall k \geq 1 \rightarrow (4)$$

TPT: $S(k+1): A_{1,k+1} = k+3$ is true.

consider $A_{1,k+1} = A_{0,p}$ (using (3)) where $p = A_{1,k} = k+2$ (using (4))

$$\begin{aligned} &= A_{0,k+2} \\ &= k+2+1 \quad (\text{using } (1)) \\ &= k+3 \quad \text{is the statement } S(k+1) \end{aligned}$$

Principles of Counting

Rules of Sum and Product:

→ The Sum Rule: If T_1 and T_2 are 2 tasks such that T_1 can be performed in 'm' different ways, T_2 can be performed in 'n' different ways and if these 2 tasks cannot be performed simultaneously, then one of the task T_1 or T_2 can be performed in $m+n$ ways.

Ex:- Suppose a library has 12 books on Maths, 10 books on physics, 16 books on Computers. Suppose the student wish to choose one of these books to study, then the no. of ways he can choose the book is $10+12+16=38$.

→ The Product Rule: If T_1 and T_2 are 2 tasks such that T_1 can be performed in 'm' different ways and for each of these ways, T_2 can be performed in 'n' different ways, then both the tasks can be performed in $m \times n$ different ways.

Ex:- Suppose a person has 8 shirts and 5 ties. Then the no. of ways of choosing a shirt and a tie is $8 \times 5 = 40$.

Problems :-

(2)

1) A license plate consists of two English letters followed by 4 digits. If repetitions are allowed, how many of the plates have only vowels and even digits?

Soln:- Each of the first 2 positions can be filled in 5 ways (with 5 vowels) & each of the remaining 4 places can be filled in 5 ways (with digits 0, 2, 4, 6, 8)

∴ No. of possible license plates of the given type is

$$(5 \times 5) \times (5 \times 5 \times 5 \times 5) = 5^6 = 15,625.$$

2) Cars of a particular company come in 4 models, 12 colors, 3 engine sizes, & 2 transmission types.

(a) How many distinct cars of this company can be manufactured?

(b) Of these, how many have the same color?

Soln:- (a) The no. of distinct cars that can be manufactured is $4 \times 12 \times 3 \times 2 = 288$.

(b) For any chosen color, no. of distinct cars that can be manufactured is

$$4 \times 3 \times 2 = 24$$

3) Find the no. of 3-digit even nos with no repeated digits.

Soln:- Consider the no. in the form xyz , where each of x, y, z represents a digit.

Since xyz has to be even $\Rightarrow z$ has to be 0, 2, 4, 6 (or) 8.

If $z=0$ then x has 9 choices & y has 8 choices.

If $z=2, 4, 6, 8$ then x has 8 choices & y has 8 choices.

[Note that y can take 0, x cannot take 0.]

∴ The desired number is $(9 \times 8 \times 1) + (8 \times 8 \times 4)$

$$= 72 + 256$$

$$= 328$$

4) A bit is either 0 or 1. A byte is a sequence of 8 bits. Find

- the no. of bytes
- the no. of bytes that begin with 11 & end with 11
- the no. of bytes that begin with 11 & do not end with 11.
- the no. of bytes that begin with 11 (or) end with 11

Soln: -

(i) Since each byte contains 8 bits & each bit is a 0 (or) 1 (two choices), the no. of bytes is $2^8 = 256$.

$\therefore$ $\downarrow \downarrow \downarrow \dots \downarrow$ constitutes a byte
 $2 \times 2 \times \dots \times 2 \text{ (8 times)} = 2^8$

(ii) In a byte beginning & ending with 11, there are 4 empty positions. These can be filled in $2^4 = 16$ ways.

$\therefore$ There are 16 bytes which begin & end with 11.

$\left[\begin{array}{c} 11 \quad \downarrow \downarrow \downarrow \downarrow \quad 11 \\ 0 \text{ (or) } 1 \end{array} \right] 2 \times 2 \times 2 \times 2 = 2^4$

(iii) In a byte beginning with 11, there are 6 empty positions. These can be filled in $2^6 = 64$ ways.

Thus there are 64 bytes that begin with 11.

Now, since there are 16 bytes that begin & end with 11, the no. of bytes that begin with 11, but do not end with 11 is $64 - 16 = 48$.

(iv) No. of bytes that end with 11 is 64 (by result (iii)).

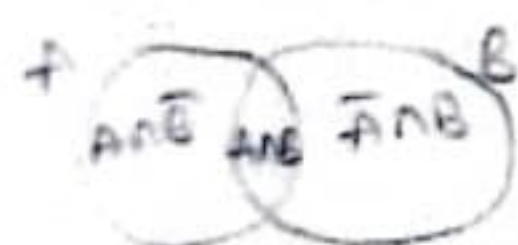
Also no. of bytes that begin & end with 11 is 16.

$\therefore$ No. of bytes that begin (or) end with 11 is

$$64 + 64 - 16 = 112. \quad \left[\begin{array}{l} \therefore \text{for any 2 finite sets } A \text{ \& } B, \\ |A \cup B| = |A| + |B| - |A \cap B| \end{array} \right]$$

(i) Begin with 11, but do not end with 11 = Begin with 11 - Begin & end with 11

(ii) Begin with 11 (or) end with 11 = Begin with 11 + end with 11 - Begin & end with 11



(iii) $\Rightarrow$ To find $A \cap B$

(iv) $\Rightarrow$ To find $A \cup B$

A : Begin with 11
 B : End with 11
 A ∩ B : Begin & end with 11
 A ∪ B : Begin (or) " " "

5) find the number of proper divisors of 441,000 ^{→ are all divisors excluding 1 & itself} (2)(a)

Soln:- we have $441000 = 2^3 \cdot 3^2 \cdot 5^3 \cdot 7^2$

∴ Every divisor of $n = 441000$ must be of the

form $d = 2^p \cdot 3^q \cdot 5^r \cdot 7^s$, where $0 \leq p \leq 3$,

$0 \leq q \leq 2$, $0 \leq r \leq 3$, $0 \leq s \leq 2$.

∴ p can be chosen in 4 ways, q in 3 ways,

r in 4 ways and s in 3 ways.

Thus total no of divisors $= 4 \times 3 \times 4 \times 3 = 144$.

but total no of proper divisors $= 144 - 2 = 142$

[- 2 ∵ a no is always divisible by 1 and by itself
so leaving these 2 choices]

$$\begin{array}{r}
 2 \overline{) 441000} \\
 2 \overline{) 220500} \\
 2 \overline{) 110250} \\
 3 \overline{) 55125} \\
 3 \overline{) 18375} \\
 5 \overline{) 6125} \\
 5 \overline{) 1225} \\
 5 \overline{) 245} \\
 7 \overline{) 49} \\
 7 \overline{) 7} \\
 1
 \end{array}$$

6) License plates consists of 2 letters followed by 4 digits.

In how many ways can this be done if

(i) No letter or digit can be repeated

(ii) Repetitions are allowed.

(iii) only vowels & Even digits can be used (with repetitions)

Soln:- (i) letter digit

Since there are 26 alphabets (letters) in English, the 1st place can be filled has 26 choices, 2nd place has 25 choices (since no repetition is allowed).

~~∴ Total no of ways it can be done is $26 \times 25 = 650$ ways~~

∵ the digits are 10 in number (0, 1, 2, ..., 9)

∴ 1st place can be filled in 10 ways, 2nd in 9 ways,

3rd in 8 ways, 4th in 7 ways.

∴ Total no of ways this can be done is $26 \times 25 \times 10 \times 9 \times 8 \times 7$
 $= 32,40,000$

$$(iii) (26 \times 26) \times (10 \times 10 \times 10 \times 10) = 26^2 \times 10^4 = 67,60,000$$

∴ repetitions are allowed.

(iii) there are 5 vowels to be used & Even digits $\Rightarrow$
0, 2, 4, 6, 8 (5 choices)

$$\therefore (5 \times 5) \times (5 \times 5 \times 5 \times 5 \times 5) = 5^6 = 15,625$$

7) Find the total no of the integers that can be formed from the digits 1, 2, 3, 4 if no digit is repeated in any one integer.

Soln: Let S_1, S_2, S_3, S_4 denote the no of integers containing 1 digit, 2 digits, 3 digits, 4 digits resp.

$S_1 \rightarrow$ 1 digit only $\Rightarrow$ 4 choices
(1 or 2 or 3 or 4)

There are 4 integers containing exactly 1 digit. $\therefore S_1 = 4$

A 2 digit integer without repetition can be chosen in $4 \times 3 = 12$ ways. $\therefore S_2 = 12$

A 3 digit integer without repetition can be chosen in $4 \times 3 \times 2 = 24$ ways. $\therefore S_3 = 24$

$$\therefore S_4 = 4 \times 3 \times 2 \times 1 \Rightarrow S_4 = 24$$

Thus the required no is $S_1 + S_2 + S_3 + S_4$
 $= 4 + 12 + 24 + 24$
 $= 64$.

8) Determine the no of 6 digit integers (no leading 0's)
in which (i) no digit is repeated (ii) No digit is repeated & it is even (iii) No digit is repeated & is divisible by 5.

Soln: (i) $\underline{\quad} \underline{\quad} \underline{\quad} \underline{\quad} \underline{\quad} \underline{\quad}$

No zero

Can be zero

There are 2 possibilities

(ii) if last place is 0 $\Rightarrow$

Care (1)

$$9 \times 9 \times 8 \times 7 \times 6 \times 5 =$$

$$9 \times 8 \times 7 \times 6 \times 5 \times 1 = S_1 \text{ (key)}$$

Case (2): if last place is not 0, 5 unit place (unit) (2) (b)
 takes 2, 4, 6, 8 (4 choices), then total no. of
 choices = $8 \times 8 \times 7 \times 6 \times 5 \times 4 = S_2 (124)$
 $\frac{2/4/6/8}{\downarrow}$
 4 choices

Thus req no. = $S_1 + S_2 =$

(iii) Case (1): If unit place is zero,
 $9 \times 8 \times 7 \times 6 \times 5 \times 1 = S_3$
 $\frac{0}{\downarrow}$
 1 choice

Case (2): If unit place is not zero, 5 ~~only~~ 5
 $8 \times 8 \times 7 \times 6 \times 5 \times 1 = S_4$
 $\frac{5}{\downarrow}$ 1 choice
 No zero, No 5 Can be zero

$\therefore$ req no. = $S_3 + S_4 =$

(2)

Permutations :- Linear arrangement of distinct objects is called Permutations.

Number of arranging 'r' distinct objects out of 'n' objects is denoted by $P(n, r)$ or ${}^n P_r$

$$\text{and } P(n, r) = n(n-1)(n-2) \dots (n-(r-1)) \\ = \frac{n!}{(n-r)!}$$

Note :- 1) No. of different arrangement of 'n' objects taken all 'n' at a time $= n!$. This is the no. of permutations of 'n' distinct objects.

2) Suppose we wish to find the no. of permutations that can be formed from a collection of 'n' objects of which n_1 are of one type, n_2 are of II type, n_3 are of III type, ... n_k are of k^{th} type and $n_1 + n_2 + \dots + n_k = n$, then no. of permutations of 'n' objects taken all at a time $= \frac{n!}{n_1! n_2! \dots n_k!}$

3) Circular arrangement of n different objects is $(n-1)!$.

Problems :- Order $\rightarrow 1, 9, 11, 7, 8, 2, 3, 5, 4, 6$

1) 5 Men and 4 women sit in a row such that the women occupy the even places. How many such arrangements are possible?

Soln :- 5 men be seated in $5!$ ways and 4 women be placed in even places in $4!$ ways.

$\therefore$ Total no. of arrangements is $5! \times 4! = 120 \times 24 = 2880$.

$$\Rightarrow n(n-1) = 90$$

$$\Rightarrow n^2 - n - 90 = 0$$

$$\Rightarrow n^2 - 10n + 9n - 90 = 0$$

$$\Rightarrow n(n-10) + 9(n-10) = 0$$

$$(n-10)(n+9) = 0$$

$$n = 10, -9$$

$$\therefore \boxed{n = 10}$$

$$(ii) P(n, 3) = 3 P(n, 2)$$

$$\frac{n!}{(n-3)!} = 3 \cdot \frac{n!}{(n-2)!}$$

$$\frac{(n-2)!}{(n-3)!} = 3$$

$$\frac{(n-2)!}{(n-3)!} = 3$$

$$\frac{(n-2)(n-3)!}{(n-3)!} = 3$$

$$\Rightarrow n-2 = 3 \Rightarrow \boxed{n = 5}$$

2) How many different strings (sequences) of length 4 can be formed using the letters of the word FLOWER?

Soln:- FLOWER has 6 letters, all of which are distinct.

The required no of strings is the no of permutations of these six letters chosen 4 at a time.

$$\text{No is } P(6, 4) = \frac{6!}{(6-4)!} = \frac{6!}{2!} = \frac{6 \times 5 \times 4 \times 3 \times 2!}{2!}$$

$$= 360.$$

3) Find the no of distinguishable permutations of the letters in the following words:

(i) BASIC

(ii) CALCULUS

(iii) MATHEMATICS

(iv) STRUCTURES

(v) ENGINEERING

Soln:- (i) The word BASIC has 5 letters, all letters being distinct.

$\therefore$ No of permutations of BASIC is $5! = 120$.

(ii) The word CALCULUS has 8 letters, of which 2 are C, 2 are L, 2 are U, and 1 each are A & S.

$\therefore$ Required No of permutations is $\frac{8!}{2! 2! 2! 1! 1!} = 5040$

(iii) The word "Mathematics" has 11 letters, in which (4)

$$\begin{array}{lll} M = 2 & H = 1 & C = 1 \\ A = 2 & E = 1 & S = 1 \\ T = 2 & I = 1 & \end{array}$$

∴ No. of permutations is $\frac{11!}{2! \times 2! \times 2! (1!)^5} = 4,989,600$

(iv) STRUCTURES has 10 letters, where

$$S = 2, T = 2, R = 2, U = 2, C = 1, E = 1$$

∴ No. of permutations is $\frac{10!}{2! \times 2! \times 2! \times 2! \times 1! \times 1!} = 2,26,800$

(v) ENGINEERING has 11 letters, where

$$E = 3, N = 3, G = 2, I = 2, R = 1$$

∴ No. of Permutations is $\frac{11!}{3! \times 3! \times 2! \times 2! \times 1!} = 2,77,200$

4) Find the no. of permutations of the letters of the word MASSASAUGA. In how many of these, all four A's are together? How many of them begin with S?

Solo:- (i) Given word has 10 letters of which 4 are A, 3 are S & 1 each are M, U, G.

∴ required no. of permutations is $\frac{10!}{4! \times 3! \times 1! \times 1! \times 1!} = 25,200$

(ii) If in a permutation, all A's are to be together, we treat all of A's as one single letter ^{say X}. Then the letters to be permuted are (AAAA), S, S, S, M, U, G, which are 7 in no.

∴ No. of Permutations is $\frac{7!}{1! \times 3! \times 1! \times 1! \times 1!} = 840$

(iii) For permutations beginning with S, there occur nine empty positions to fill, where two are S, 4 are A, 1 each are M, U, G.

$$\therefore \text{No. of such permutations} = \frac{9!}{2! 4! (1!)^3} = 7560.$$

5) Find the no. of permutations of the letters of the word MISSISSIPPI. How many of these begin with ^{an} I? How many of these begin ^{& end} with a S? ~~How many of these have no consecutive S's?~~

Soln:- (i) Given word has 11 letters, in which there are 4 I's, 4 S's, 2 P's, 1 M.

$$\therefore \text{No. of permutations is } \frac{11!}{4! 4! 2! 1!} = 34,650$$

(ii) For permutations beginning with I, there occur 10 empty positions to fill, where 3 are I, 4 are S, 2 are P, 1 M.

$$\therefore \text{No. of such permutations is } \frac{10!}{3! 4! 2! 1!} = 12,600$$

(iii) For permutations beginning & ending with S, there occur 9 empty positions to fill, where 2 are S, 4 are I, 2 are P, 1 M.

$$\therefore \text{No. of such permutations} = \frac{9!}{2! 4! 2! 1!} = 3780.$$

(iv) ~~Let all 4 S be treated as a single letter X, so that there are 1 M, 4 I, 2 P & 1 X, totalling to 8. These can be arranged in~~

6) (a) How many arrangements are there for all letters in the word SOCIOLOGICAL?

(b) In how many of these arrangements (i) A & G are adjacent (ii) all the vowels are adjacent.

~~$\frac{8!}{1! 4! 2! 1!} = 840 \text{ ways. Also the 4 S's can be arranged among themselves in } 4! \text{ ways. } \therefore \text{req. no.} = 840 \times 4! = 20,160.$~~

Sol:- (a) Given word has 12 letters, of which 3 are o, (5)
2 each are C, I, L & 1 each are S, A, G.

∴ no of arrangements of these letters is

$$\frac{12!}{3! 2! 2! 2! (1!)^3} = 99,79,200.$$

(b) (i) If in an arrangement, A & G are to be adjacent, we treat A & G ^{together} as a single letter say X, such ~~are~~ ^{that} there are 3 o's, 2 C's, 2 I's, 2 L's, 1 S & 1 X, totalling to 11 letters. These can be arranged in

$$\frac{11!}{3! 2! 2! 2! 1! 1!} = 8,31,600 \text{ ways}$$

Also, the letters A & G can be arranged among themselves in 2 ways [∵ A & G → total 2, each is 1, 1

$$\Rightarrow \frac{2!}{1! 1!} = 2]$$

∴ total no of arrangements in this case is

$$8,31,600 \times 2 = 1,663,200$$

(ii) If in an arrangement, all the vowels are to be adjacent, we treat all the vowels present in the given word (namely A, o, I) as a single letter, say Y, such that there are 2 C's, 2 L's, 1 S, 1 G, 1 Y, totalling to 7 letters. These can be arranged in

$$\frac{7!}{2! 2! 1! 1! 1!} = 1,260 \text{ ways.}$$

Also, since the given word contains 3 o's, 2 I's & 1 A, the letters o, I, A (clubbed as Y) can be arranged among themselves ^{totalling to 6} in

$$\frac{6!}{3! 2! 1!} = 60 \text{ ways.}$$

∴ Total no of arrangements in this case is

$$1260 \times 60 = 75,600$$

7) How many six digit numbers can one make using the digits 1, 3, 3, 7, 7, 8 ?

Soln:- Here $n=6$.

The given digits are 1, 3, 3, 7, 7, 8 in which there are

1 1's, 2 3's, 2 7's, 1 8.

↓ ↓ ↓ ↓
1 type 2nd type 3rd type 4th type

∴ No of six digits no's using given digits is

$$= \frac{6!}{1! 2! 2! 1!}$$

$$= \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{1 \times 2 \times 2 \times 1}$$

$$= 30 \times 6$$

$$= 180$$

8) How many the integers 'n' can we form using the digits 3, 4, 4, 5, 5, 6, 7 if we want 'n' to exceed 50,00,000

Soln:- Let $n = x_1 x_2 x_3 x_4 x_5 x_6 x_7$

where $x_1, x_2, \dots, x_7$ are the given digits.

Since n exceeds 50,00,000, x_1 can take 5 (or) 6 (or) 7

If $x_1 = 5$, then $x_2 \dots x_7$ is an arrangement of the remaining 6 digits which contains 2 4's & one each are 3, 5, 6, 7.

∴ No of such arrangements is $\frac{6!}{2! 1! 1! 1! 1! 1!} = 360$.

If $x_1 = 6$, then $x_2 \dots x_7$ is an arrangement of the remaining 6 digits which contains 2 4's, 2 5's, and one each are 3, 7.

∴ No of such arrangements = $\frac{6!}{2! 2! 1! 1! 1! 1!} = 180$.

If $x_1 = 7$, then $x_2 \dots x_7$ is an arrangement of the remaining 6 digits which contains 2 4's, 2 5's, 1 each of 3, 6. (6)

$\therefore$ No. of such arrangements is $\frac{6!}{2! 2! 1! 1!} = 180$.

$\therefore$ By sum Rule, the no. of 'n's of the desired type is

$$360 + 180 + 180 = 720.$$

9) In how many ways can 6 men & 6 women be seated in a row

(i) if any person may sit next to each other?

(ii) if men & women must occupy alternate seats?

Soln: (i) If any person may sit next to each other, no distinction be made b/w men & women in their seating.

$\therefore$ Number of ways they can be seated is $12! [(6+6)!]$
 $= 479,001,600$

(ii) If men & women are to occupy alternate seats, 6 men can be seated in $6!$ ways in odd places & 6 women " " " " even places.

$\therefore$ No. of ways in which men occupy ~~even~~ odd places & women occupy even places is $6! \times 6! = 518400$.

Similarly No. of ways in which men occupy even places & women occupy odd places is $6! \times 6! = 518400$

$\therefore$ Total no. of ways men & women occupy alternate seats is $518400 + 518400 = 10,36,800$.

10) for non-negative integers n & r , if $n+1 > r$, prove that

$$P(n+1, r) = \frac{(n+1)}{(n+1-r)} P(n, r).$$

Soln:- we have $P(n, r) = \frac{n!}{(n-r)!}$ & $P(n+1, r) = \frac{(n+1)!}{(n+1-r)!}$

$$\begin{aligned} \therefore \frac{P(n+1, r)}{P(n, r)} &= \frac{(n+1)!}{(n+1-r)!} \times \frac{(n-r)!}{n!} \\ &= \frac{(n+1)!}{n!} \times \frac{(n-r)!}{(n+1-r)!} \\ &= \frac{(n+1) \cdot n!}{n!} \times \frac{(n-r)!}{(n+1-r)(n-r)!} \\ &= \frac{(n+1)}{(n+1-r)} \end{aligned}$$

Thus $P(n+1, r) = \left(\frac{n+1}{n+1-r} \right) P(n, r)$.

- 11) 4 different Mathematics books, 5 different CO books & 2 different GTC books are to be arranged in a shelf. How many different arrangements are possible if
- the books in each particular subject must all be together?
 - only the mathematics books must be together?

Soln:- (a) The mathematics books can be arranged among themselves in $4!$ ways, the CO books can be arranged among themselves in $5!$ ways. the GTC books can be arranged among themselves in $2!$ ways and the 3 groups in $3!$ ways.

$\therefore$ No of possible arrangements = $4! \times 5! \times 2! \times 3! = 34,560$

(b) Since the mathematics books has to be together, Consider the 4 mathematics books as one single book.

Then we have 8 books, which can be arranged in $8!$ ways. But the mathematics books can be arranged among themselves in $4!$ ways.

$\therefore$ the required no of arrangements is $8! \times 4! = 9,67,680$

Combinations: Selecting a set of 'r' objects from a set of 'n' objects ($n \geq r$) without regard to order is called a combination of 'r' objects.

The total no. of combination of 'r' different objects that can be selected from 'n' different objects is denoted by $C(n, r)$ or nC_r and is given by ${}^nC_r = \frac{n!}{(n-r)! r!}$

Note:- ${}^nC_r = {}^nC_{n-r}$

Problems:-

- 1) How many committees of 5 members with one chair person can be selected from 12 persons?

Soln:- 1 chair person out of 12 can be selected in 12 ways. Remaining 4 members out of 11 persons can be selected in ${}^{11}C_4$ ways.

$$\therefore \text{Req. no. of committees} = 12 \times {}^{11}C_4 = 3960.$$

- 2) A certain Q.P contains 3 parts A, B, C with 4 questions in Part A, 5 in Part B, 6 in Part C. It is required to answer 7 questions selecting atleast 2 from each part. In how many different ways can a student select his questions for answering?

Soln:-

	No. of questions it contains	No. of questions chosen		
		case (1)	case (2)	case (3)
A	4	2	2	3
B	5	2	3	2
C	6	3	2	2
No. of ways of selection		${}^4C_2 \times {}^5C_2 \times {}^6C_3$ (S_1)	${}^4C_2 \times {}^5C_3 \times {}^6C_2$ (S_2)	${}^4C_3 \times {}^5C_2 \times {}^6C_2$ (S_3)

$$\therefore \text{Total no. of ways} = S_1 + S_2 + S_3 = 1200 + 900 + 600 = 2700.$$

3) Find the no. of committees of 5 that can be selected from 7 men and 5 women, if the committee is to consist of atleast 1 man and atleast 1 woman?

Soln:-

Men	Women	No. of committees of 5
1	4	${}^7C_1 \times {}^5C_4 = 35$
2	3	${}^7C_2 \times {}^5C_3 = 210$
3	2	${}^7C_3 \times {}^5C_2 = 350$
4	1	${}^7C_4 \times {}^5C_1 = 175$

$\therefore$ Req. no. of committees containing ^{atleast} 1 man and 1 woman is

$$35 + 210 + 350 + 175 = 770$$

4) A woman has 11 close relatives and she wishes to invite 5 of them to dinner. In how many ways can she invite them in the following situations:

- i) there is no restriction on the choice
- ii) 2 particular persons will not attend separately.
- iii) 2 " " " " together.

Soln:- i) Since there is no restriction on the choice, 5 out of 11 can be invited in ${}^{11}C_5$ ways = 462 ways.

ii) Let A and B be 2 ^{particular} persons. The following cases arises:

- a) If both A and B are invited, then no. of ways of selection is ${}^9C_3 = 84$ ways.
- b) If both A and B are not invited, then no. of ways of selection is ${}^9C_5 = 126$ ways.

$$\therefore \text{Total no. of ways} = 84 + 126 = 210.$$

iii) The following cases arises:

- a) If A is invited and B is not invited, then no. of ways of selection is ${}^9C_4 = 126$ ways.
- b) If B is invited and A is not invited, then no. of ways of selection is ${}^9C_4 = 126$ ways.

c) If both A and B are not invited, then no. of ways of selection is ${}^9C_5 = 126$ ways. (2)

$$\text{Total no. of ways} = 126 + 126 + 126 \\ = 378.$$

5) Find the no. of arrangements of all the letters of the word TALLAHASSEE. How many of these arrangements have no adjacent A's?

Soln:- i) Given word has 11 letters, in which there are 3 A's, 2 each are L's, S's, E's and 1 each are T and H.

$$\therefore \text{No. of such arrangements is } \frac{11!}{3! (2!)^3 (1!)^2} = 8,21,600.$$

ii) If the A's are disregarded, the remaining 8 letters can be arranged in $\frac{8!}{(2!)^3 (1!)^2} = 5040$ ways.

In each of these arrangements, there are 9 possible locations for the 3 A's. $[\rightarrow T \rightarrow L \rightarrow L \rightarrow H \rightarrow S \rightarrow S \rightarrow E \rightarrow E]$ These locations can be chosen in 9C_3 ways.

$$\therefore \text{No. of arrangements having no adjacent A's is } 5040 \times {}^9C_3 \\ = 4,23,360.$$

6) Find how many distinct 4-digit integers one can make from the digits 1, 3, 3, 7, 7, 8?

Soln:- The following 3 cases arises:

i) 4-digit integers containing all distinct no.'s (1, 3, 7, 8) can be done in $4!$ ways = 24 ways.

ii) 4-digit integers containing 2 pairs of identical no.'s (3, 3, 7, 7) can be done in $\frac{4!}{2! 2!} = 6$ ways.

P.T.O.

iii) 4-digit integers containing 1 pair of identical no.'s are:

3, 3, 1, 7 (or) 3, 3, 1, 8 (or) 3, 3, 7, 8 (or)

7, 7, 1, 3 (or) 7, 7, 1, 8 (or) 7, 7, 3, 8

which are 6 in no.

This can be done in $\frac{4!}{2!1!1!} \times 6 = 72$ ways.

$\therefore$ Total no. of ways = $24 + 6 + 72 = 102$

7) A man has 15 close friends of whom 6 are women. In how many ways can he invite 3 or more of these friends to a party if he wants the same no. of men (including himself) as women?

soln: out of 15 friends, 6 are women and 9 are men.

women	men	No. of ways
2	1 (+ himself in all cases)	$6C_2 \times 9C_1 = 135$
3	2	$6C_3 \times 9C_2 = 720$
4	3	$6C_4 \times 9C_3 = 1260$
5	4	$6C_5 \times 9C_4 = 756$
6	5	$6C_6 \times 9C_5 = 126$

$\therefore$ Total no. of ways = 2997.

8) How many bytes contain i) exactly two 1's ii) exactly four 1's iii) exactly 6 1's iv) atleast six 1's.

soln:- 1 byte = 8 bits and each bit is a 0 or 1.

i) Two 1's can be chosen in $8C_2$ ways and each of the remaining 6 locations are filled by zeros only (1 choice).
 $\therefore$ No. of bytes containing exactly two 1's = $8C_2 \times 1^6 = 28$

ii) $8C_4 \times 1^4 = 70$.

iii) $8C_6 \times 1^2 = 28$.

iv) atleast 6 1's can be chosen in the following ways:

$$6 \text{ 1's} \Rightarrow {}^8C_6$$

$$7 \text{ 1's} \Rightarrow {}^8C_7$$

$$8 \text{ 1's} \Rightarrow {}^8C_8$$

$\therefore$ No. of bytes with atleast 6 1's is

$${}^8C_6 + {}^8C_7 + {}^8C_8 = 37.$$

9) Prove the following identities:

$$i) C(n, r-1) + C(n, r) = C(n+1, r).$$

$$ii) C(m, 2) + C(n, 2) = C(m+n, 2) - mn.$$

Soln:- i) $C(n, r-1) + C(n, r) = {}^nC_{r-1} + {}^nC_r$

$$= \frac{n!}{(n-r+1)!(r-1)!} + \frac{n!}{(n-r)!r!}$$

$$= \frac{n!}{(n-r+1)(n-r)!(r-1)!} + \frac{n!}{(n-r)!r(r-1)!}$$

$$= \frac{n!}{(n-r)!(r-1)!} \left[\frac{1}{n-r+1} + \frac{1}{r} \right]$$

$$= \frac{n!}{(n-r)!(r-1)!} \left[\frac{r + n-r+1}{r(n-r+1)} \right]$$

$$= \frac{n!}{(n-r)!(r-1)!} \times \frac{(n+1)}{r(n-r+1)}$$

$$= \frac{(n+1)!}{r!(n-r+1)!} = C(n+1, r).$$

$$ii) C(m, 2) + C(n, 2) = \frac{m!}{(m-2)!2!} + \frac{n!}{(n-2)!2!}$$

$$= \frac{m(m-1)(m-2)!}{2(m-2)!} + \frac{n(n-1)(n-2)!}{2(n-2)!}$$

$$= \frac{1}{2} [m(m-1) + n(n-1)]$$

$$= \frac{1}{2} [m^2 + n^2 - m - n]$$

$$= \frac{1}{2} [(m+n)^2 - 2mn - m - n]$$

$$= \frac{1}{2} [(m+n)^2 - (m+n) - 2mn]$$

$$= \frac{1}{2} [(m+n)(m+n-1)] - mn$$

$$= \frac{1}{2} \left[\frac{(m+n)(m+n-1)(m+n-2)!}{(m+n-2)!} \right] - mn$$

$$= \frac{1}{2!} \frac{(m+n)!}{(m+n-2)!} - mn$$

$$= c(m+n, 2) - mn.$$

Binomial and Multinomial Theorem :

pg 19, 20, 21
not there (14)

A basic property of $C(n, r)$ is that, it is the coefficient of $x^{n-r} y^r$ in the expansion of the expression $(x+y)^n$, where x & y are any two real no's. In other words, the expansion of $(x+y)^n$ in powers of x & y is as follows

$$\begin{aligned}(x+y)^n &= x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_{n-1} x y^{n-1} + y^n \\ &= \sum_{r=0}^n {}^nC_r x^{n-r} y^r \longrightarrow (1).\end{aligned}$$

Since ${}^nC_r = {}^nC_{n-r}$;

$$(x+y)^n = \sum_{r=0}^n {}^nC_r x^r y^{n-r} \longrightarrow (2)$$

Thus nC_r is the coefficient of $x^r y^{n-r}$ in the expansion of $(x+y)^n$.

nC_r is also denoted by $\binom{n}{r}$

$$\begin{aligned}\therefore \textcircled{1} \Rightarrow (x+y)^n &= \sum_{r=0}^n \binom{n}{r} x^r y^{n-r} \\ \textcircled{2} \Rightarrow (x+y)^n &= \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r\end{aligned} \quad \textcircled{3}$$

Expression $\textcircled{3}$ is called Binomial theorem for a +ve integral index 'n'.

The no's $\binom{n}{r}$ for $r = 0, 1, 2, \dots, n$, namely $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \dots, \binom{n}{n}$

in the above result are called the Binomial Coefficients.

The generalization of Binomial Theorem is known as

Multinomial Theorem :

Statement :— For +ve integers n & k , the coefficient of $x_1^{n_1} x_2^{n_2} \dots x_k^{n_k}$ in the expansion of $(x_1 + x_2 + \dots + x_k)^n$ is

$$\frac{n!}{n_1! n_2! \dots n_k!}$$

where each $n_i \leq n$ & $n_1 + n_2 + \dots + n_k = n$.

Note :-

1) The general term in the expansion of $(x_1 + x_2 + \dots + x_k)^n$ is

$$\frac{n!}{n_1! n_2! \dots n_k!} x_1^{n_1} x_2^{n_2} \dots x_k^{n_k} \quad \left(\text{Alternate statement of Multinomial theorem} \right)$$

3) The expression $\frac{n!}{n_1! n_2! \dots n_k!}$ is also written as $\binom{n}{n_1, n_2, \dots, n_k}$

3) Multinomial theorem can also be stated as

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{n_i} \binom{n}{n_1, n_2, \dots, n_k} x_1^{n_1} x_2^{n_2} \dots x_k^{n_k}$$

Problems :-

1) Prove the following identities for a true integer n :

$$(i) \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = 2^n$$

$$(ii) \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n} = 0$$

Soln :- The Binomial theorem for a true integral index n is given by

$$(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}$$

(i) Taking $x=1$; $y=1$, we get

$$2^n = \sum_{r=0}^n \binom{n}{r} 1^r 1^{n-r} = \sum_{r=0}^n \binom{n}{r}$$

$$\therefore 2^n = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n}$$

(ii) Taking $x=-1$; $y=1$, we get

$$0 = \sum_{r=0}^n \binom{n}{r} (-1)^r 1^{n-r} = \sum_{r=0}^n (-1)^r \binom{n}{r}$$

$$\Rightarrow 0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n}$$

P.T.O

2) Compute the following. (15)

(i) $\binom{12}{5,3,2,2}$

(ii) $\binom{8}{4,2,2,0}$

(iii) $\binom{10}{5,3,2,2}$

Soln:- (i) $\binom{12}{5,3,2,2} = \frac{12!}{5!3!2!2!} = 166,320$

(ii) $\binom{8}{4,2,2,0} = \frac{8!}{4!2!2!} = 420$

(iii) Meaningless since $5+3+2+2 = 12 > n=10$

3) Find the sum of all coefficients in the expansion of

(a) $(x+y)^n$

(b) $(x_1+x_2+\dots+x_k)^n$

Soln:- (a) we have $(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^r y^{n-r}$

Taking $x=1; y=1$, we get

$$2^n = \sum_{r=0}^n \binom{n}{r} 1^r 1^{n-r} = \sum_{r=0}^n \binom{n}{r}$$

$$\text{i.e. } 2^n = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n}$$

Thus, the sum of all the coefficients in the expansion of $(x+y)^n$ is 2^n .

(b) By Multinomial theorem, we have

$$\sum_{n_1} \binom{n}{n_1, n_2, \dots, n_k} x_1^{n_1} x_2^{n_2} \dots x_k^{n_k} = (x_1 + x_2 + \dots + x_k)^n$$

Taking $x_1=1, x_2=1, \dots, x_k=1$, we get

$$\sum_{n_1} \binom{n}{n_1, n_2, \dots, n_k} = (1+1+\dots+1)_{k \text{ times}}^n = k^n$$

Thus, the sum of all the coefficients in the expansion of $(x_1+x_2+\dots+x_k)^n$ is k^n .

4) Obtain the sum of all the coefficients in the expansion of

(i) $(x+y)^{10}$

(ii) $(x+y+z)^{12}$

(iii) $(x+y+z+w)^5$

(iv) $(2s-3t+5u+6v-11w+3x+2y)^{10}$

Soln:- (i) From result (a) of above problem, the sum of the coefficients in the expansion of $(x+y)^{10}$ is 2^{10} .

(ii) From result (b) of above problem, the sum of the coefficients in the expansion of $(x+y+z)^{12}$ is 3^{12} .

(iii) From result (b) again, sum of coeff in the expansion of $(x+y+z+w)^5$ is 4^5 .

(iv) By virtue of the multinomial theorem, we have

$$\begin{aligned} (2s - 3t + 5u + 6v - 11w + 3x + 2y)^{10} &= \\ &= \sum_{n_i} \binom{10}{n_1, n_2, \dots, n_7} (2s)^{n_1} (-3t)^{n_2} (5u)^{n_3} (6v)^{n_4} (-11w)^{n_5} \\ &\quad (3x)^{n_6} (2y)^{n_7} \\ &= \sum_{n_i} \binom{10}{n_1, n_2, \dots, n_7} 2^{n_1} (-3)^{n_2} 5^{n_3} 6^{n_4} (-11)^{n_5} 3^{n_6} 2^{n_7} s^{n_1} t^{n_2} u^{n_3} v^{n_4} w^{n_5} x^{n_6} y^{n_7} \end{aligned}$$

Taking $s=1, t=1, u=1, v=1, w=1, x=1, y=1$ we get

$$(2 - 3 + 5 + 6 - 11 + 3 + 2)^{10} = \sum_{n_i} \binom{10}{n_1, n_2, \dots, n_7} 2^{n_1} (-3)^{n_2} 5^{n_3} 6^{n_4} (-11)^{n_5} 3^{n_6} 2^{n_7}$$

The sum in the RHS of the above expression is the sum of the coefficients in the expansion of $(2s - 3t + 5u + 6v - 11w + 3x + 2y)^{10}$.

$$\begin{aligned} \text{This sum} &= (2 - 3 + 5 + 6 - 11 + 3 + 2)^{10} \\ &= 4^{10} \end{aligned}$$

5) Find the coefficient of

(i) $x^3 y^9$ in the expansion of $(x+2y)^{12}$.

(ii) x^0 in the expansion of $(3x^2 - \frac{2}{x})^{15}$

(iii) x^{12} " " " $x^3 (1-2x)^{10}$

(iv) x^k " " " $(1+x+x^2)(1+x)^n$, where

n is a true integer & $0 \leq k \leq n+2$.

Soln (i) By Binomial theorem,

(16)

$$(x+2y)^{12} = \sum_{r=0}^{12} \binom{12}{r} x^r (2y)^{12-r}$$
$$= \sum_{r=0}^{12} \binom{12}{r} 2^{12-r} \cdot x^r \cdot y^{12-r}$$

Taking $r=3$ in the above expansion, the coefficient of $x^3 y^9$ is $\binom{12}{3} 2^9 = \frac{12!}{9! 3!} \cdot 2^9 = 1,126,400$

(ii) By Binomial theorem,

$$\left(3x^2 - \frac{2}{x}\right)^{15} = \sum_{r=0}^{15} \binom{15}{r} (3x^2)^r \cdot \left(-\frac{2}{x}\right)^{15-r}$$
$$= \sum_{r=0}^{15} \binom{15}{r} 3^r \cdot (-2)^{15-r} \cdot x^{2r} \cdot \left(\frac{1}{x}\right)^{15-r}$$
$$= \sum_{r=0}^{15} \binom{15}{r} 3^r \cdot (-2)^{15-r} \cdot x^{2r} \cdot x^{r-15}$$
$$= \sum_{r=0}^{15} \binom{15}{r} 3^r \cdot (-2)^{15-r} \cdot x^{3r-15}$$

Taking $r=5$ in the above expansion, the coefficient of x^0 is $\binom{15}{5} 3^5 \cdot (-2)^{10} = {}^{15}C_5 \cdot 3^5 \cdot (-2)^{10}$
 $= 74,724,2496$

(iii) By Binomial theorem,

$$(1-2x)^{10} = \sum_{r=0}^{10} \binom{10}{r} 1^{10-r} \cdot (-2x)^r$$
$$\therefore x^3 (1-2x)^{10} = \sum_{r=0}^{10} \binom{10}{r} \cdot (-2)^r \cdot x^{r+3}$$

Taking $r=9$ in the above expansion, we get the coeff of x^{12} as $\binom{10}{9} (-2)^9 = {}^{10}C_9 \cdot (-2)^9$
 $= -5,120$

(iv) Using binomial theorem,

$$(1+x)^n = \sum_{r=0}^n \binom{n}{r} 1^{n-r} x^r$$

$$\therefore (1+x+x^2)(1+x)^n = (1+x+x^2) \sum_{r=0}^n \binom{n}{r} x^r$$

$$= \sum_{r=0}^n \binom{n}{r} x^r + \sum_{r=0}^n \binom{n}{r} x^{r+1} + \sum_{r=0}^n \binom{n}{r} x^{r+2}$$

Coeff of x^k is obtained by putting $r=k$ in 1st sum, $r=k-1$ in 2nd sum & $r=k-2$ in 3rd sum

ie coeff of x^k is

$$\binom{n}{k} + \binom{n}{k-1} + \binom{n}{k-2}$$

Say; for ex:- coeff of x^{15} is $\binom{n}{15} + \binom{n}{14} + \binom{n}{13}$

6) Determine the coefficient of the following :-

(i) xyz^5 in the expansion of $(x+y+z)^7$

(ii) $x^2y^2z^3$ " " " $(3x-2y-4z)^7$

(iii) x^4y^4 " " " $(2x^3-3xy^2+z^2)^6$

(iv) xyz^{-2} " " " $(x-2y+3z^{-1})^4$

(v) $a^2b^3c^2d^5$ " " " $(a+2b-3c+2d+5)^{16}$

Soln:- (i) By multinomial theorem, we have that the General term in the expansion of $(x+y+z)^7$ is

$$\binom{7}{n_1, n_2, n_3} x^{n_1} y^{n_2} z^{n_3}$$

for $n_1=1, n_2=1, n_3=5$, we have

$$\binom{7}{1, 1, 5} xyz^5 = \frac{7!}{1! 1! 5!} xyz^5 = 42 xyz^5$$

This shows that the required coefficient of xyz^5 is 42.

(ii) The general term in the expansion of $(3x - 2y - 4z)^7$ (17)

$$\text{is } \binom{7}{n_1, n_2, n_3} (3x)^{n_1} (-2y)^{n_2} (-4z)^{n_3}$$

for $n_1 = 2$; $n_2 = 2$, $n_3 = 3$, we have

$$\begin{aligned} \binom{7}{2, 2, 3} (3x)^2 (-2y)^2 (-4z)^3 \\ = \binom{7}{2, 2, 3} 3^2 \cdot (-2)^2 \cdot (-4)^3 x^2 y^2 z^3 \end{aligned}$$

$\therefore$ coeff of $x^2 y^2 z^3$ is $\binom{7}{2, 2, 3} \times 9 \times 4 \times (-64)$

$$= -2304 \times \frac{7!}{2! 2! 3!} = -4,83,840$$

(iii) The general term in the expansion of $(2x^3 - 3xy^2 + z^2)^6$ is

$$\binom{6}{n_1, n_2, n_3} (2x^3)^{n_1} (-3xy^2)^{n_2} (z^2)^{n_3}$$

$$= \binom{6}{n_1, n_2, n_3} 2^{n_1} (-3)^{n_2} \cdot x^{3n_1} \cdot x^{n_2} \cdot y^{2n_2} \cdot z^{2n_3}$$

$$= \binom{6}{n_1, n_2, n_3} 2^{n_1} (-3)^{n_2} \cdot x^{3n_1 + n_2} \cdot y^{2n_2} \cdot z^{2n_3}$$

$$\begin{aligned} 3n_1 + n_2 &= 11 \\ 3n_1 + n_2 &= 9 \\ n_1 &= 3 \end{aligned}$$

for $n_3 = 0$; $n_2 = 2$; $n_1 = 3$, we have

$$\binom{6}{3, 2, 0} 2^3 (-3)^2 \cdot x^{11} y^4$$

$\therefore$ coeff of $x^{11} y^4$ is $2^3 (-3)^2 \cdot \binom{6}{3, 2, 0}$

$$= 8 \times 9 \times \frac{6!}{3! 2! 0!} = 4,320$$

(iv) The general term in the expansion of $(x - 2xy + 3z^{-1})^4$ is

$$\binom{4}{n_1, n_2, n_3} x^{n_1} (-2y)^{n_2} (3z^{-1})^{n_3}$$

$$= \binom{4}{n_1, n_2, n_3} (-2)^{n_2} 3^{n_3} \cdot x^{n_1} \cdot y^{n_2} \cdot z^{-n_3}$$

for $n_1=1, n_2=1, n_3=2$, we have

$$\binom{4}{1,1,2} (-2)^1 \cdot 3^2 \cdot xyz^{-2}$$

$$\therefore \text{coeff of } xyz^{-2} \text{ is } -2 \cdot 9 \times \binom{4}{1,1,2}$$

$$= -18 \times \frac{4!}{1!1!2!} = -216$$

(v) The general term in the expansion of $(a+2b-3c+2d+5)^{16}$ is $\binom{16}{n_1, n_2, n_3, n_4, n_5} a^{n_1} (2b)^{n_2} (-3c)^{n_3} (2d)^{n_4} 5^{n_5}$

for $n_1=2, n_2=3, n_3=2, n_4=5$; $n_5 = 16 - (2+3+2+5)$
 i.e. $n_5=4$, we have $\left[\begin{array}{l} \because n_1 + \dots + n_5 = n \\ \Rightarrow n_5 = n - (n_1 + \dots + n_4) \end{array} \right]$

$$\binom{16}{2,3,2,5,4} 2^3 \cdot (-3)^2 \cdot 2^5 \cdot 5^4 a^2 b^3 c^2 d^5$$

$$\therefore \text{coeff of } a^2 b^3 c^2 d^5 \text{ is } 8 \times 9 \times 2^5 \cdot 5^4 \times \binom{16}{2,3,2,5,4}$$

$$= 1440000 \times \frac{16!}{2!3!2!5!4!}$$

Ans
 Find the coefficient of:

1) $x^5 y^2$ in the expansion of $(2x-3y)^7$

2) $x^3 z^4$ " " " $(x+y+z)^7$

3) $x^3 y^3 z^2$ " " " $(2x-3y+5z)^8$

4) $w^3 x^2 y z^2$ " " " $(2w-x+3y-2z)^8$

5) $x_1^2 x_3 x_4^3 x_5^4$ " " " $(x_1+x_2+x_3+x_4+x_5)^{10}$

Combinations with Repetitions:

(18)

Suppose we have to select a combination of r -objects, with repetitions, from a set of n -distinct objects, where $r \leq n$, then the no. of such selections is given by

$$C(n+r-1, r) = \frac{(n+r-1)!}{r! (n-1)!} = C(r+n-1, n-1)$$

i.e. $C(n+r-1, r) = C(r+n-1, n-1)$ represents the no. of

Combinations of n distinct objects, taken ' r ' at a time, with repetitions allowed.

Note:- $\Rightarrow$ Eqⁿ (1) represents the no. of ways in which r identical objects can be distributed among n distinct containers.

0, 1, 2, 3, ...

2) Eqⁿ (1) also represents the no. of non-negative integer soln's of the Eqⁿ $x_1 + x_2 + \dots + x_n = r$.

[A non-negative integer soln of the Eqⁿ $x_1 + x_2 + \dots + x_n = r$ is an n -tuple $(x_1, x_2, \dots, x_n)$, where $x_1, x_2, \dots, x_n$ are non-negative integers whose sum is r]

$n \geq r$ - combⁿ — out of n , we select r

Problems:-

1) In how many ways can we distribute 10 identical marbles among 6 distinct containers?

places $\rightarrow n$
dist. item $\rightarrow r$

Soln:- Here $r = 10$; $n = 6$.

$\therefore$ Required no. is $C(6+10-1, 10) = C(15, 10)$

$$= \frac{15!}{10! 5!} = 3003.$$

2) Find the no. of ways of placing 8 identical balls in 5 numbered boxes?

Soln:- Here $r = 8$; $n = 5$

Required no. is $C(5+8-1, 8) = C(12, 8) = \frac{12!}{8! 4!} = 495.$

3) A bag contains coins of 7 different denominations, with at least 1 dozen coins in each denomination. In how many ways can we select a dozen coins from the bag?

Soln: Here $r=12$; $n=7$.

The no of ways making a selection of dozen coins from the bag is $C(7+12-1, 12) = C(18, 12) = \frac{18!}{12!6!} = 18,564$.

4) Find the no of non-negative integer solutions of the eqⁿ
 $x_1 + x_2 + x_3 + x_4 + x_5 = 8$.

Soln: The required no is $C(5+8-1, 8) = C(12, 8)$
 $\left[\text{Since here } r=8; n=5 \right] = \frac{12!}{8!4!} = 495$.

5) Find the no of non-negative integer solⁿ of the eqⁿ
 $x_1 + x_2 + x_3 + x_4 = 7$

Soln: Here $n=4$; $r=7$.

Required no is $C(4+7-1, 7) = C(10, 7) = \frac{10!}{7!3!}$
 $=$

6) Find the no of distinct terms in the expansion of
 $(x_1 + x_2 + x_3 + x_4 + x_5)^{16}$

Soln: - The general term in the expansion of $(x_1 + x_2 + x_3 + x_4 + x_5)^{16}$ is of the form $\binom{16}{n_1, n_2, n_3, n_4, n_5} x_1^{n_1} x_2^{n_2} x_3^{n_3} x_4^{n_4} x_5^{n_5}$ where each n_i is a non-negative integer & the sum of these n_i 's is 16.

$\therefore$ no of distinct terms in the expansion = the no of non-negative integer solutions of the eqⁿ $n_1 + n_2 + n_3 + n_4 + n_5 = 16$

$\therefore$ Required no is $C(5+16-1, 16) = C(20, 16) = 4845$.

7) Find the no of non-negative integer soln's of the inequality $x_1 + x_2 + \dots + x_6 < 10$.

(19)

Soln:- Given $x_1 + x_2 + \dots + x_6 < 10$
 $\Rightarrow x_1 + x_2 + \dots + x_6 = 9 - x_7$, $\left[\begin{array}{l} \text{ex. } 2+3 < 10 \\ \Rightarrow 2+3 = 9-4 \end{array} \right]$

so that x_7 is a non-negative integer.

$\therefore$ the required no is the no of non-negative soln's of the eqn $x_1 + x_2 + \dots + x_7 = 9$.

This no is $C(7+9-1, 9) = (15, 9) = 5005$.

8) Find the no of distinct terms in the expansion of $(w+x+y+z)^{10}$

Soln:- The general term in the expansion of $(w+x+y+z)^{10}$ is $\binom{10}{n_1, n_2, n_3, n_4} w^{n_1} x^{n_2} y^{n_3} z^{n_4}$ where each n_i is a non-negative integer & the sum of these n_i 's is 10.

$\therefore$ no of distinct terms in the expansion = no of non-negative integer soln of the eqn $n_1 + n_2 + n_3 + n_4 = 10$

This no = $C(4+10-1, 10) = C(13, 10) = 286$.

9) Find the no of non-negative integer soln of the inequality $x_1 + x_2 + \dots + x_5 \leq 19$

Soln:- Given $x_1 + x_2 + \dots + x_5 \leq 19$
 $\Rightarrow x_1 + x_2 + \dots + x_5 = 19 - x_6$, so that x_6 is non-negative

$\therefore$ Required no = no of non-negative integer solns of the eqn $x_1 + x_2 + \dots + x_5 + x_6 = 19$

This no is $C(6+19-1, 19) = C(24, 19) = 42,504$.

10) Find the no. of integer solutions of
 $x_1 + x_2 + \dots + x_5 = 30$ where $x_1 \geq 2, x_2 \geq 3, x_3 \geq 4,$
 $x_4 \geq 2, x_5 \geq 0.$ $\rightarrow (1)$

Soln:- Let $y_1 = x_1 - 2; y_2 = x_2 - 3; y_3 = x_3 - 4$
 $y_4 = x_4 - 2; y_5 = x_5.$

then $y_1, y_2, \dots, y_5$ are all non-negative integers.

Given Eqⁿ now becomes,

$$(y_1 + 2) + (y_2 + 3) + (y_3 + 4) + (y_4 + 2) + y_5 = 30$$

$$\Rightarrow y_1 + y_2 + y_3 + y_4 + y_5 = 30 - 11$$

$$\Rightarrow y_1 + \dots + y_5 = 19. \rightarrow (2)$$

$y_i \geq 0$ for $i = 1, 2, \dots, 5$

No. of non-negative integer soln of (2) is

$$C(5 + 19 - 1, 19) = C(24 - 1, 19) = C(23, 19) \\ = 8855$$

11) How many integer soln's are there to $x_1 + x_2 + \dots + x_5 = 20$
tho where $x_i \geq 2$?

Soln:- Given $x_1 \geq 2; x_2 \geq 2; \dots; x_5 \geq 2.$

Let $y_1 = x_1 - 2; y_2 = x_2 - 2; \dots; y_5 = x_5 - 2$

$\therefore$ Given Eqⁿ becomes

$$y_1 + 2 + y_2 + 2 + y_3 + 2 + y_4 + 2 + y_5 + 2 = 20.$$

$$\Rightarrow y_1 + y_2 + \dots + y_5 = 10.$$

$$\therefore \text{Required } \underline{\text{no.}} = C(5 + 10 - 1, 10) = C(14, 10) = 1001.$$

12) In how many ways can we distribute 12 identical pencils
tho to 5 children so that every child gets atleast 1 pencil?

Soln:- First let us distribute one pencil to each child.

So, ^{that} Remaining 7 pencils are to be distributed to 5 children. This can be ^{done} in

$$C(5 + 7 - 1, 7) = C(11, 7)$$

$$= 330 \text{ ways.}$$

$$n = 5, \\ r = 7$$

13) In how many ways can 10 identical pens be distributed among 5 children in the following cases:-

(i) There are no restrictions.

(ii) Each child gets at least one pen

(iii) The youngest child gets at least 2 pens

Soln:- (i) Since there are no restrictions in distributing 10 pens to 5 children, no of ways of distribution is $C(5+10-1, 10) = C(14, 10) = 1001$

(ii) First we distribute one pen to each child. Then the remaining 5 pens are to be distributed among 5 children in $C(5+5-1, 5) = C(9, 5) = 126$ ways.

(iii) First we give 2 pens to the youngest child, then the remaining 8 pens are to be distributed among 5 children. This can be done in $C(5+8-1, 8) = C(12, 8) = 495$ ways.

14) In how many ways can one distribute eight identical balls into 4 distinct containers so that

(i) No container is empty?

(ii) the 4th container gets an odd no of balls?

Soln:- (i) First we distribute one ball to each container. Then the remaining 4 balls are to be distributed among 4 containers & this can be done in $C(4+4-1, 4) = C(7, 4) = 35$ ways

(ii) If the 4th container has to get an odd no of balls, we have to put 1 (or) 3 (or) 5 (or) 7 balls into it.

Case (i):- Suppose we put 1 ball to 4th container, then the remaining 7 balls ~~is to~~ ^{can} be distributed into the remaining 3 containers in $C(3+7-1, 7) = C(9, 7)$ ways.

Case (ii):- Suppose we put 3 balls to 4th container, then the remaining 5 balls can be distributed into the remaining

3 containers in $C(3+5-1, 5) = C(7, 5)$ ways.

Case (iii) $\therefore$ Suppose we put 5 balls to 4^{th} container, then the remaining 3 balls can be distributed into the remaining 3 containers in $C(3+3-1, 3) = C(5, 3)$ ways.

Case (iv) $\therefore$ $C(3+1-1, 1) = C(3, 1)$ ways.

$\therefore$ total no. of ways of distributing the given balls with the given condⁿ is

$$C(9, 7) + C(7, 5) + C(5, 3) + C(3, 1) = 70.$$

15) Find the no. of ways of distributing 7 identical pens & 7 identical pencils to 5 children so that each gets at least 1 pen & at least 1 pencil.

Soln $\therefore$ First let us distribute 1 pen & 1 pencil to each child, then the remaining 2 pens can be distributed to 5 children in $C(5+2-1, 2) = C(6, 2)$ ways.

Similarly the remaining 2 pencils can be distributed to 5 children in $C(5+2-1, 2) = C(6, 2)$ ways.

$\therefore$ no. of ways of distributing the given sets under the given condⁿ is $C(6, 2) \times C(6, 2)$

$$= 225$$

16) Find the no. of ways of giving 10 identical gift boxes to 6 persons A, B, C, D, E, F in such a way that the total no. of boxes given to A & B together does not exceed 4.

Soln $\therefore$ of the 10 boxes, suppose 'x' boxes are given to A & B together, then $0 \leq x \leq 4$.

$\therefore$ no. of ways of giving 'x' boxes to A & B is

$$C(2+x-1, x) = C(x+1, x) = \frac{(x+1)!}{x! 1!} = \frac{(x+1)x!}{x!} = x+1 \rightarrow (1).$$

P.T.O.

no. of ways in which remaining $(10-r)$ boxes can be given to C, D, E, F is

(21)

$$C(4 + (10-r) - 1, (10-r)) = C(13-r, 10-r)$$

→ (2).

Consequently, the no. of ways in which 'r' boxes may be given to A & B and $(10-r)$ boxes to C, D, E, F is

$$(r+1) \times C(13-r, 10-r).$$

Since $0 \leq r \leq 4$, Total no. of ways in which the boxes may be given is,

$$\sum_{r=0}^4 (r+1) \times C(13-r, 10-r).$$

- 17) A total amount of Rs 1500 is to be distributed to 3 poor students A, B, C of a class. In how many ways the distribution can be made in multiples of Rs 100 if
- (i) everyone of these must get atleast Rs 300?
 - (ii) A must get atleast Rs 500, & B & C must get atleast Rs 400 each?

Soln:- Taking Rs 100 as a unit, there are 15 units for distribution, among 3 students A, B, C.

Case (i): Each of the 3 students must get atleast 3 units. $\therefore$ let us first distribute 3 units to each A, B, C. then the remaining 6 units are to be distributed to A, B, C, and this can be done in $C(3+6-1, 6) = C(8, 6) = 28$ ways.

Case (ii): A must get atleast 5 units & B & C must get atleast 4 units each. $\therefore$ let us first distribute 5 units to A & 4 units each to B & C, then the remaining 2 units are to be distributed to A, B, C, This can be done in $C(3+2-1, 2) = C(4, 2) = 6$ ways.

18) A total of Rs 10,000 is to be distributed to 4 persons A, B, C, D in multiples of Rs 1000. In how many ways can the distribution be done (i) if there is no restriction (ii) if everyone of these persons should receive at least Rs 1000? (iii) If everyone should receive at least Rs 1000 & A in particular should receive at least Rs 5000?

Sol:- Taking Rs 1000 as one unit, there are 10 units for distribution, among 4 persons.

Case (i):- Since there is no restriction, 10 units can be distributed among 4 persons in

$$C(4+10-1, 10) = C(13, 10) = 286 \text{ ways.}$$

Case (ii):- Each of the 4 students must get at least 1 unit. Let us first distribute 1 unit to each A, B, C, D. Then the remaining 6 units are to be distributed among 4 persons. This can be done in

$$C(4+6-1, 6) = C(9, 6) = 84 \text{ ways.}$$

Case (iii):- A should receive ^{at least} 5 units & each of B, C, D should receive at least 1 unit. ~~then the~~

Let us first distribute 5 units to A, and 1 unit each to B, C, D. Then the remaining 2 units are to be distributed among 4 persons. This can be done in

$$C(4+2-1, 2) = C(5, 2) = 10 \text{ ways.}$$

19) A message is made up of 12 different symbols & is to be transmitted through a communication channel. In addition to the 12 symbols, the transmitter will also send a total of 45 blank spaces b/w the symbols, with at least 3 spaces b/w each pair of consecutive symbols. In how many ways can the transmitter send such a message?

Soln:- The 12 symbols can be arranged in $12!$ ways. (22)

For each of these arrangements, there are 11 positions

b/w the 12 symbols. Since there must be at least 3 spaces

b/w ~~consecutive~~ ^{we can make use of 33 blank spaces out} symbols, ~~33 of the 45 spaces will be~~
~~used up~~. The remaining $(45-33)=12$ ^{blank} spaces can be accommodated

in 11 positions. This can be done in $C(11+12-1, 12)$

$= C(22, 12)$ ways.

$\therefore$ No of ways the transmitter can send such a message

$$\text{is } C(22, 12) \times 12! = \frac{22!}{10! 12!} \times 12! = \frac{22!}{10!} = 3.0974 \times 10^{14}$$

20) Six distinct symbols are transmitted through a communication channel. A total of 12 blanks are to be inserted b/w the symbols with at least 2 blanks b/w every pair of symbols. In how many ways can the symbols and the blanks be arranged?

Soln:- The 6 symbols can be arranged in $6!$ ways. For each of these arrangements, there are 5 positions b/w the 6 symbols. Since there must be at least 2 blanks b/w every pair of symbols, 10 of the 12 blanks will be used up. The remaining 2 blanks are to be accommodated in 5 positions. This can be done in $C(5+2-1, 2) = C(6, 2)$ ways.

$\therefore$ No of ways the symbols & the blanks can be arranged is $C(6, 2) \times 6! = 10,800$.